Hybrid-PDP: Simultaneously Enforcing both Record-level and User-level Personalized Differential Privacy

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ABSTRACT

Traditional differential privacy (DP) offers two levels of privacy protection across a dataset, namely *record-level DP* and *user-level DP*. Both of these protections are applied uniformly across records or users, failing to accommodate the heterogeneous privacy requirements that arise in practice. To address this limitation, personalized differential privacy (PDP) was introduced to support non-uniform privacy budgets. However, existing work focuses on either record-level PDP or user-level PDP, without enabling simultaneous personalized protection at both levels. In practice, such hybrid privacy protection is essential under the PDP setting: assigning personalized privacy budgets disrupts the traditional dominance of user-level protection over individual records, particularly in realistic scenarios where users contribute multiple records and datasets may be aggregated from multiple parties. Motivated by these limitations, we introduce *hybrid personalized differential privacy* (Hybrid-PDP), a new privacy model that simultaneously enforces personalized protections at both the record- and user-levels. We develop a general framework for achieving Hybrid-PDP, built upon two novel primitives. The first, *prefix-truncation*, breaks the bidirectional privacy dependencies between records and users, while the second, *doubly-truncation*, mitigates the stringent constraints imposed by user-level personalization. Extensive experimental results demonstrate the high utility of our framework. For sum and count queries, our approach achieves less than 3% relative error on most real-world datasets under meaningful privacy settings, yielding orders-of-magnitude improvements over straightforward Hybrid-PDP baselines that rely on predefined global parameters.

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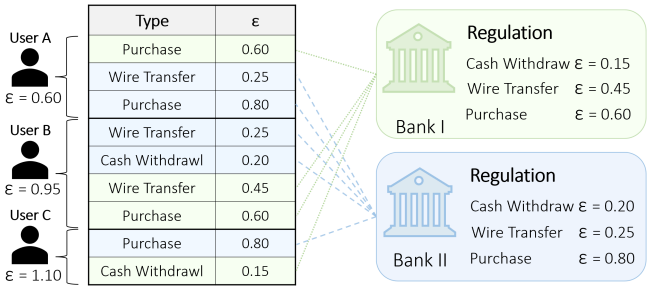


Figure 1: A bank example under Hybrid-PDP.

1 INTRODUCTION

Differential Privacy (DP) [12] provides a rigorous mathematical framework to protect individuals' information in private data releases. It has been widely adopted across the industry [6, 15, 35] and national statistical agencies [26]. Roughly speaking, DP ensures that the output distribution remains ε-indistinguishable whether any single individual's data is included or excluded, where ε is called the privacy budget and controls the level of protection: a smaller ε provides a stronger privacy guarantee. Standard DP adopts two different policies: *record-DP* [13], which protects individual data records; *user-DP* [2, 8, 10, 17], which protects all of a user's data records as a whole. Consider a banking scenario where a dataset consists of financial records of a bank's customers, such as their transactions, deposits, and credit card repayments. Record-level DP protects each individual financial record, whereas user-level DP protects all information corresponding to a single customer. It is trivial to see that, with the same privacy budget, user-DP provides stronger privacy protection.

Although such standard DP policies are widely adopted, the assumption of uniform privacy protection for all records or users may not be desirable. Following the banking example, it is common that different customers have their own privacy budgets due to policy considerations associated with their occupations, and different categories of financial records also naturally fall under different privacy regulations or internal policies across different banks. For example, a casual coffee purchase transaction may require less protection than a large cash deposit. To accommodate such diverse privacy requirements, *personalized DP* (PDP) was proposed [1, 14, 18], which allows personalized privacy budgets across a dataset. Similar to the standard DP model, current PDP research also adopts two levels of privacy protections: *record-PDP* [4, 18, 24] and *user-PDP* [34]. Under these models, each record or user can have its own privacy budget to better reflect real-world needs.

However, a limitation of these two PDP models is that neither can simultaneously provide personalized protection at both the record- and user-levels. Unlike in the standard DP setting, where user-level DP strictly dominates record-level DP, there is no such dominance relationship between user-PDP and record-PDP. Intuitively, this is because when both records and users have their own personalized privacy budgets, and each user can own multiple records, we cannot guarantee that user-level privacy protection always dominates the record-level one. This is especially true when the data are contributed by multiple parties, each with their own privacy policies. For example, as shown in Figure 1, when a dataset contains financial records from multiple banks, each bank may have different privacy regulations for different categories of financial records. In this case, a single bank cannot determine a customer's personalized privacy budget to sufficiently protect all of that customer's financial data across all banks.

Motivated by these limitations, we introduce *hybrid personalized DP* (Hybrid-PDP), which provides personalized privacy protection at both the user and record levels simultaneously. Let $\mathcal{U}$ and $\mathcal{R}$ denote the domains of users and records, respectively. We define a record-level privacy budget function $\Phi_{\text{rec}} : \mathcal{R} \to [\check{\varepsilon}_{\text{rec}}, \hat{\varepsilon}_{\text{rec}}]$ and a user-level privacy budget function $\Phi_{\text{user}} : \mathcal{U} \to [\check{\varepsilon}_{\text{user}}, \hat{\varepsilon}_{\text{user}}]$, where $[\check{\varepsilon}_{\text{rec}}, \hat{\varepsilon}_{\text{rec}}]$ and $[\check{\varepsilon}_{\text{user}}, \hat{\varepsilon}_{\text{user}}]$ denote the predefined ranges of record- and user-level privacy budgets, respectively. A mechanism satisfies $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP if its output distribution is simultaneously $\Phi_{\text{rec}}(r)$-indistinguishable under the addition or removal of any record $r \in D$, and $\Phi_{\text{user}}(u)$-indistinguishable under the addition or removal of any subset of records belonging to any user $u \in D$, including all records of that user. Notably, under this definition, both privacy budget functions and the domains of users and records are public information, whereas the presence or absence of any particular user or record in the dataset is hidden and protected. This is consistent with existing notions of user-PDP [34] and record-PDP [4, 18], while jointly accounting for both levels of protection.

One straightforward approach here is to use a uniform privacy budget at the record-level, which can be derived from some global bounds. More specifically, we use the smaller of $\check{\varepsilon}_{\text{rec}}$ and $\check{\varepsilon}_{\text{user}}/\hat{m}$ where $\hat{m}$ is the largest number of records that can be owned by a user. However, this approach inevitably introduces an unacceptable error that is tied to global bounds, which are extreme values that typically do not occur in most datasets. A more desirable error should be independent of such global bounds. Instead, we would like to have an "instance-specific" error, which depends on $\varepsilon_{\min}(D) = \min\{\varepsilon^{\text{rec}}_{\min}(D), \min_{u \in D} \frac{\Phi_{\text{user}}(u)}{m(D,u)}\}$, where $\varepsilon^{\text{rec}}_{\min}(D)$ and $\min_{u \in D} \frac{\Phi_{\text{user}}(u)}{m(D,u)}$ can be regarded as instance-specific versions of $\check{\varepsilon}_{\text{rec}}$ and $\check{\varepsilon}_{\text{user}}/\hat{m}$ (see Section 4.2 for the formal definitions of $\varepsilon_{\min}(D)$ and $m(D,u)$). Indeed, reducing the error dependency from global parameters to instance-specific ones is also a key goal of prior record-PDP [4] and user-PDP [34] works.

However, achieving such instance-specific error bounds is highly challenging. A natural idea is to estimate the above parameters under both record-PDP and user-PDP. Yet this approach inevitably violates either record-level or user-level privacy requirements, due to the "bidirectional" privacy dependency between records and users. More precisely, running a record-PDP mechanism to estimate certain record-level parameters also incurs a corresponding privacy loss for the user who owns those records. Symmetrically, running a user-PDP mechanism to estimate user-level parameters inevitably induces additional privacy loss for the individual records associated with that user. In addition to the above challenges, a solution for this new hybrid privacy model should be general to cater to all query types. This ultimately leads to the main research problem of this work:

> *"Does there exist a general Hybrid-PDP solution that can be used to answer arbitrary queries while achieving an instance-specific error bound?"*

1.1 Our Results

In this paper, we give a positive answer to the above research problem by presenting a novel and general framework that transforms any existing record-DP mechanism to support Hybrid-PDP. To address the bidirectional privacy dependency problem, we first develop *prefix-truncation*, in which we delete records for each user until the cumulative privacy budget of the retained records falls below that user's privacy budget. Here, we prioritize deleting the records with larger privacy budgets. This ensures that the insertion or deletion of a single record in the initial dataset affects the truncated result by at most two records. After this process, record-level privacy dominates the user-level privacy, so that we can invoke a record-PDP mechanism to realize the Hybrid-PDP model. However, prefix-truncation alone cannot handle cases where user-level privacy constraints are relatively more stringent than those at the record-level, as this may force us to truncate too many records. As a result, in the worst case, the error introduced by prefix-truncation can even exceed that of the naive global-bound method. To address this issue, we further introduce *doubly-truncation*, where, given a threshold τ, we first delete records whose privacy budgets fall below this threshold and shrink the remaining privacy budgets to τ before applying prefix-truncation. After the doubly-truncation, all retained records have a privacy budget of τ, and each user's privacy constraint is dominated by that of the records corresponding to the user. Thus, we can invoke a record-DP mechanism with this budget to satisfy Hybrid-PDP. This resolves the utility issue of prefix-truncation by choosing a good threshold τ. Finally, we complete this story by further proposing a Laplace-based threshold selection mechanism with a carefully calibrated score function to find a good τ. With this idea, we present a general framework such that, for any query Q, given a record-DP mechanism, we construct a corresponding Hybrid-PDP solution whose error depends on $\varepsilon_{\min}(D)$.

In summary, our contributions are threefold:

- To the best of our knowledge, this is the first work to formally present the Hybrid-PDP model, which enables heterogeneous personalized privacy requirements at different levels simultaneously, better addressing real-world needs.
- As our main result, we present a general framework for arbitrary Hybrid-PDP queries. Given a record-DP mechanism $\mathcal{M}$ with an error of $\text{Err}_{\mathcal{M}}(D, \varepsilon_{\min}(D))$, our framework achieves an error bounded by $\tilde{O}(\text{Err}_{\mathcal{M}}(D, \varepsilon_{\min}(D)))$ [1], with

[1] $\tilde{O}(\cdot)$ denotes asymptotic notation that suppresses polylogarithmic factors.

a running time of $\tilde{O}(M+N)$. Here, M and N denote the number of users and records in D, respectively.

- We conduct a comprehensive empirical study on the practical performance of our proposed framework across real-world and synthetic datasets. The results highlight that our framework provides up to 600× utility improvement while maintaining efficiency.

2 RELATED WORK

Providing personalized privacy protection is a timely research area that is gaining significant momentum, driven by heterogeneous privacy requirements in real-world applications [21]. Notably, the concept of personalized privacy was explored even before the introduction of DP: for example, Xiao and Tao [38] studied it within the framework of k-anonymity, which was later extended to other privacy models [16, 32, 39].

Under the rigorous DP model [11, 13], which has become a dominant notion for privacy, the idea of personalized privacy has been extensively explored, both in theory [1, 4, 14, 18, 23, 28, 29, 34] and in real-world applications [3, 24, 25, 27, 36, 40, 42]. These models are referred to as PDP models. Although the terms "records" and "users" are often used interchangeably, or even imprecisely, in many studies, we emphasize a fundamental distinction: in user-level PDP, the personalized privacy budget is assigned to a group of records as a whole, whereas in record-level PDP, the budget is assigned directly to individual records. Thus, prior work can be broadly categorized into two main streams of PDP models: the user-level PDP model [34] and the record-level PDP model [4, 18, 23, 24, 30, 31]. However, existing PDP models provide personalized privacy protection at only one level at a time. This work is the first to simultaneously enforce personalized privacy requirements at both the record and user levels. We further develop a practical and general solution that achieves Hybrid-PDP with high utility and computational efficiency, distinguishing our work from prior PDP studies.

3 PRELIMINARY

3.1 Notation

Let $\mathcal{U}$ denote the possibly infinite domain of users. For each user $u \in \mathcal{U}$, let $\mathcal{R}_u$ denote the possibly infinite domain of records associated with u. The global record domain is then $\mathcal{R} = \bigcup_{u\in\mathcal{U}} \mathcal{R}_u$. Given a dataset D, for any user $u \in \mathcal{U}$, let $D(u) \subseteq \mathcal{R}_u$ denote the set of records contributed by u to D. Let $m(D,u) = |D(u)|$ denote the number of records owned by user u in D. Then, $D = \bigcup_{u\in\mathcal{U}} D(u)$. Moreover, we write $u \in D$ if user u contributes at least one record to D, i.e., $m(D,u) > 0$.

For two datasets D and D', the *record-level distance* $d_{\text{rec}}(D,D')$ is the minimum number of record insertions or deletions required to transform D into D'. We say D and D' are r-neighbors (denoted $D \sim_r D'$) if $d_{\text{rec}}(D,D') = 1$ and they differ by record $r \in \mathcal{R}$. Similarly, the *user-level distance* $d_{\text{user}}(D,D')$ is the minimum number of user-level insertions or deletions required to transform D into D', where each user-level operation inserts or deletes all or any subset of records associated with a single user. We say D and D' are u-neighbors (denoted $D \sim_u D'$) if $d_{\text{user}}(D,D') = 1$, meaning the two datasets differ only in the records associated with a certain user u. Furthermore, for clarity of notation, we use $\widetilde{val}$ to denote the privacy-preserving estimation of any true value val.

3.2 Differential Privacy

Definition 3.1 (Differential privacy). *For any $\varepsilon > 0$, a randomized mechanism $\mathcal{M}$ with output space $\mathcal{Y}$ satisfies ε-differential privacy (DP) if, for any pair of neighboring datasets $D \sim D'$ and any possible output set $Y \subseteq \mathcal{Y}$ of $\mathcal{M}$,*

$$\Pr\big[\mathcal{M}(D) \in Y\big] \le e^{\varepsilon} \Pr\big[\mathcal{M}(D') \in Y\big].$$

By integrating user-neighbors and record-neighbors, we can achieve record-level or user-level privacy protection, referred to as *record-DP* and *user-DP*, respectively. DP satisfies several desirable properties, as outlined below.

Lemma 1 (Group privacy). *If a mechanism $\mathcal{M}$ with output space $\mathcal{Y}$ satisfies ε-DP, then for any two datasets D and D' with $d(D,D') = \lambda$ (i.e., differing by λ records or users), and any possible output set $Y \subseteq \mathcal{Y}$ of $\mathcal{M}$, the following holds:*

$$\Pr\big[\mathcal{M}(D) \in Y\big] \le e^{\lambda\varepsilon} \Pr\big[\mathcal{M}(D') \in Y\big].$$

Lemma 2 (Post-Processing). *If a mechanism $\mathcal{M}$ is ε-DP, then for any mechanism $\mathcal{M}'$, the composed output $\mathcal{M}'(\mathcal{M}(D))$ is still ε-DP.*

Lemma 3 (Sequential Composition [13]). *Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be two randomized mechanisms, where $\mathcal{M}_1$ satisfies ε_1-DP and $\mathcal{M}_2$ satisfies ε_2-DP. Then, $\mathcal{M}(D) = \mathcal{M}_2(D, \mathcal{M}_1(D))$ preserves $(\varepsilon_1+\varepsilon_2)$-DP.*

A commonly used DP mechanism is the *Laplace mechanism*. Given any query $Q : \mathcal{D} \to \mathbb{R}$, its global sensitivity GS_Q is defined as $\text{GS}_Q = \max_{D\sim D'} |Q(D) - Q(D')|$.

Lemma 4 (Laplace Mechanism [13]). *Given a query $Q : \mathcal{D} \to \mathbb{R}$, the Laplace mechanism $\mathcal{M}(D) = Q(D) + \eta$ satisfies ε-DP, where η is drawn from a Laplace distribution with scale GS_Q/ε.*

Note that all the aforementioned properties and mechanisms are applicable to both record-DP and user-DP by adopting the corresponding notations under a specific DP policy. For example, under the record-DP setting, we employ the record-level distance $d_{\text{rec}}(D,D')$ in Lemma 1 and the record-neighboring relation in Lemma 2 and Lemma 3.

3.3 Personalized Differential Privacy

Standard DP assigns a uniform privacy budget to all units, e.g., records or users. However, this assumption is often unrealistic. For example, in a banking scenario, confidential transfers and routine transactions may require different levels of privacy protection.

To accommodate such heterogeneous privacy requirements, prior work proposed *personalized differential privacy* (PDP) [18], which assigns different privacy budgets to different records or users. Under PDP, we have a privacy function Φ to assign the privacy budget for a unit x, which can be either a user or a record.

Definition 3.2 (Personalized Differential Privacy (PDP)). *For any privacy function Φ, a randomized mechanism $\mathcal{M}$ with output space $\mathcal{Y}$ satisfies Φ-PDP if, for any pair of neighboring datasets $D \sim_x D'$ and any possible output set $Y \subseteq \mathcal{Y}$ of $\mathcal{M}$, the following holds:*

$$\Pr\big[\mathcal{M}(D) \in Y\big] \le e^{\Phi(x)} \Pr\big[\mathcal{M}(D') \in Y\big].$$

Similar to standard DP, PDP also supports both record-level and user-level privacy protection. By letting x denote either a user u or a record r, we obtain *user-PDP* and *record-PDP*, respectively.

Limitations of PDP. Although PDP supports heterogeneous privacy, it can be applied only at a single level, e.g., either user-level or record-level. However, in practice, both users and records may simultaneously require personalized protection. In the banking example, each customer may have different privacy requirements due to occupation-related policies: some occupations require stricter confidentiality, while others allow higher disclosure risk. Meanwhile, different transactions may exhibit varying levels of privacy sensitivity, influenced by factors such as the merchant category, associated banking policies, or time of occurrence, which are independent of customer-level policies. These observations motivate the introduction of a novel PDP model that satisfies two-level heterogeneous privacy requirements.

Indeed, this privacy issue, requiring user- and record-level privacy guarantees simultaneously, is unique to the PDP setting and does not arise in standard DP. In standard DP, user-level DP always dominates record-level DP: an ε-user-DP mechanism naturally provides at least ε-record-DP, since adding or removing a user also covers the addition or removal of any individual record owned by that user. However, PDP introduces additional challenges. Records may have different privacy budgets, and the user-record ownership relation, that is, whether a record associated with a given user exists in the dataset, is itself private. As a result, user-PDP does not necessarily dominate record-PDP.

4 HYBRID PERSONALIZED DIFFERENTIAL PRIVACY

Motivated by the limitations of existing PDP frameworks, we propose a novel privacy model called *Hybrid Personalized Differential Privacy* (Hybrid-PDP), which provides heterogeneous privacy protection at both the record- and user-levels. The detailed mathematical formulation of Hybrid-PDP and its desired properties are presented in Section 4.1. In Section 4.2, we introduce a straightforward idea to achieve Hybrid-PDP for arbitrary queries. However, such an idea can offer only limited utility, as we elaborate later.

4.1 Definition and Basic Properties

To provide both user- and record-level privacy protection simultaneously, we define two public privacy functions, $\Phi_{\text{rec}} : \mathcal{R} \to [\check{\varepsilon}_{\text{rec}}, \hat{\varepsilon}_{\text{rec}}]$ and $\Phi_{\text{user}} : \mathcal{U} \to [\check{\varepsilon}_{\text{user}}, \hat{\varepsilon}_{\text{user}}]$, respectively, where $[\check{\varepsilon}_{\text{rec}}, \hat{\varepsilon}_{\text{rec}}]$ and $[\check{\varepsilon}_{\text{user}}, \hat{\varepsilon}_{\text{user}}]$ are predefined ranges with conservatively small lower bounds and large upper bounds, respectively. More precisely, each user u is assigned a privacy budget of $\Phi_{\text{user}}(u)$, and each record r is assigned a privacy budget of $\Phi_{\text{rec}}(r)$. For later use, we also define $\hat{\varepsilon} = \min(\hat{\varepsilon}_{\text{rec}}, \hat{\varepsilon}_{\text{user}})$ and $\check{\varepsilon} = \min(\check{\varepsilon}_{\text{rec}}, \check{\varepsilon}_{\text{user}}/\hat{m})$, where $\hat{m}$ denotes a public finite contribution bound such that each user can contribute at most $\hat{m}$ records to any dataset. Our Hybrid-PDP model can be formalized as:

Definition 4.1 (Hybrid Personalized Differential Privacy). *A randomized mechanism $\mathcal{M}$ with output space $\mathcal{Y}$ satisfies $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP if both of the following conditions hold for any possible output set $Y \subseteq \mathcal{Y}$ of $\mathcal{M}$:*

- ***Record-level constraint.*** *For every record $r \in \mathcal{R}$ and every pair of r-neighboring datasets $D \sim_r D'$,*
$$\Pr\big[\mathcal{M}(D) \in Y\big] \le e^{\Phi_{\text{rec}}(r)} \Pr\big[\mathcal{M}(D') \in Y\big].$$
- ***User-level constraint.*** *For every user $u \in \mathcal{U}$ and every pair of u-neighboring datasets $D \sim_u D'$,*
$$\Pr\big[\mathcal{M}(D) \in Y\big] \le e^{\Phi_{\text{user}}(u)} \Pr\big[\mathcal{M}(D') \in Y\big].$$

Remark. Notably, both the privacy functions and the domains of users and records are public. Thus, for any user u or record r, the values $\Phi_{\text{rec}}(r)$ and $\Phi_{\text{user}}(u)$ are public. In contrast, whether u or r exists in the dataset D remains private. As a result, the privacy budgets associated with the users and records present in D remain private. This is consistent with the privacy model adopted in existing PDP works [4, 18, 34].

Lemma 5 (Hybrid-PDP Post-processing). *If a mechanism $\mathcal{M}$ is $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP, then for any mechanism $\mathcal{M}'$, the composed output $\mathcal{M}'(\mathcal{M}(D))$ is still $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP.*

Lemma 6 (Hybrid-PDP Sequential Composition). *Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be two randomized mechanisms such that $\mathcal{M}_1$ satisfies $(\Phi^1_{\text{rec}}, \Phi^1_{\text{user}})$-Hybrid-PDP and $\mathcal{M}_2$ satisfies $(\Phi^2_{\text{rec}}, \Phi^2_{\text{user}})$-Hybrid-PDP. Then, $\mathcal{M}(D) = \mathcal{M}_2(D, \mathcal{M}_1(D))$ satisfies $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP, where for all $r \in \mathcal{R}$ and $u \in \mathcal{U}$, $\Phi_{\text{rec}}(r) = \Phi^1_{\text{rec}}(r) + \Phi^2_{\text{rec}}(r)$, $\Phi_{\text{user}}(u) = \Phi^1_{\text{user}}(u) + \Phi^2_{\text{user}}(u)$.*

The proofs of the two lemmas above are similar to the corresponding proofs for standard DP [13]. Additionally, to facilitate later analysis, we introduce the following instance-wise extremal quantities under the Hybrid-PDP framework. For a dataset D with user- and record-level privacy functions $(\Phi_{\text{user}}, \Phi_{\text{rec}})$, define

$$\begin{aligned} \varepsilon^{\text{user}}_{\min}(D) &:= \min_{u \in D} \Phi_{\text{user}}(u), \\ m_{\max}(D) &:= \max_{u \in D} m(D, u), \\ \varepsilon^{\text{rec}}_{\min}(D) &:= \min_{r \in D} \Phi_{\text{rec}}(r). \end{aligned}$$

Notably, since the presence or absence of records and users is private, all the above parameters are private and should be protected.

4.2 Straw-man: Utilizing Global Bounds of Privacy Parameters to Achieve Hybrid-PDP

Here, we propose a straightforward baseline mechanism for achieving $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP. The high-level idea is to reduce the Hybrid-PDP problem to a record-DP problem, enabling the use of existing record-level mechanisms to answer the query. The key challenge lies in deriving a uniform record-level privacy budget that simultaneously ensures the two-level protection guarantee of Hybrid-PDP.

Naively, such a uniform privacy budget can be derived from the predefined global parameters $\check{\varepsilon}_{\text{rec}}$, $\check{\varepsilon}_{\text{user}}$ and $\hat{m}$. Specifically, we use a uniform global record-level budget $\check{\varepsilon} = \min(\check{\varepsilon}_{\text{rec}}, \check{\varepsilon}_{\text{user}}/\hat{m})$. It is straightforward to show that this approach naturally satisfies both the record-level and user-level privacy requirements defined by the $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP guarantee.

(i) *Record-level privacy guarantee.* For any $r \in \mathcal{R}$, we have $\check{\varepsilon} \le \Phi_{\text{rec}}(r)$. Thus, this mechanism is $\check{\varepsilon}$-DP at the record level, and therefore satisfies the record-level privacy requirement of Φ_{rec}.

(ii) *User-level privacy guarantee.* For any $u \in \mathcal{U}$, we have $\hat{m}\check{\varepsilon} \leq \Phi_{\text{user}}(u)$. Since any user can own at most $\hat{m}$ records, by the group privacy property (Lemma 1), the total privacy loss of any user is bounded by $\hat{m}\check{\varepsilon}$. Thus, it satisfies the user-level privacy requirement specified by Φ_{user}.

Moving toward the utility analysis, recall that for any query Q, we reduce the Hybrid-PDP problem to a record-DP problem, and a record-DP mechanism $\mathcal{M}$ is involved. Here, we first formalize the error of $\mathcal{M}$ as $\text{Err}_{\mathcal{M}}(D, \varepsilon, \beta)$: for a given dataset D, privacy budget ε, with probability at least $1-\beta$, we have $|Q(D) - \mathcal{M}(D)| \leq \text{Err}_{\mathcal{M}}(D, \varepsilon, \beta)$. Then, we have

Lemma 7. *For any $\beta \in [0, 1)$, privacy budget functions Φ_{rec} and Φ_{user}, and dataset D, by applying a record-DP mechanism $\mathcal{M}$ with error $\text{Err}_{\mathcal{M}}(\cdot)$ as defined above and setting $\check{\varepsilon}$ as the privacy budget, the naive baseline achieves, with probability at least $1-\beta$, an error of $\text{Err}_{\mathcal{M}}(D, \check{\varepsilon}, \beta)$.*

Utility limitation. As we have shown, $\check{\varepsilon}$ is solely determined by the predefined global parameters rather than the specific data instance. This is often undesirable, as the global parameters tend to be overly conservative in order to cover all privacy requirements across the record and user universe. It leads to significant utility degradation, as most existing standard DP mechanisms have error inversely proportional to the privacy budget, which in our case becomes $1/\check{\varepsilon}$.

Instead, a more desirable error should have some instance-specific dependency. We would like to achieve an error of

$$\text{Err}_{\mathcal{M}}(D, c \cdot \varepsilon_{\min}(D), \beta), \quad \beta \in [0, 1) \tag{1}$$

where c is a constant and $\varepsilon_{\min}(D)$ is defined as

$$\varepsilon_{\min}(D) \;:=\; \min\left\{\varepsilon^{\text{rec}}_{\min}(D),\ \min_{u \in D} \frac{\Phi_{\text{user}}(u)}{m(D,u)}\right\}. \tag{2}$$

Roughly speaking, $\varepsilon_{\min}(D)$ reflects the most stringent actual privacy requirement, whether user- or record-level, present in the queried dataset D. Thus, it leads to an instance-specific error for D.

To provide intuition on how this improves utility, consider the bank example. The globally strongest privacy requirements for certain transactions or customers may be extremely small, resulting in $\check{\varepsilon} \sim 10^{-9}$. However, the actual value of $\varepsilon_{\min}(D)$ in a specific bank database is often orders of magnitude larger, as such sensitive transactions or customers may not appear in that bank.

5 A HYBRID-PDP FRAMEWORK WITH INSTANCE-SPECIFIC ERROR

In this section, we study how to achieve instance-specific error under Hybrid-PDP. In Section 5.1, we present two straightforward attempts. Although these attempts fail to satisfy Hybrid-PDP, they provide the foundation for our final solution. The difficulty arises from privacy dependencies between records and users, which prevent us from processing records and users independently. To eliminate these privacy dependencies, in Section 5.2 we introduce a *prefix-truncation* mechanism. However, as discussed in Section 5.3, prefix-truncation alone cannot achieve the desired error bound. This motivates a more comprehensive doubly-truncation mechanism, which addresses the limitation by upgrading the privacy budgets of certain records. Next, in Section 5.4, we show how doubly-truncation can serve as a building block for a threshold-selection algorithm under Hybrid-PDP. By designing a private scoring function for each candidate threshold, the algorithm selects a threshold that balances the trade-off between truncating too many records and using privacy budgets that are too small, which leads to excessive noise. Finally, in Section 5.5, building on the idea of prefix-truncation, we propose our Hybrid-PDP solution that achieves instance-specific errors for arbitrary queries.

5.1 Straightforward Attempts

Attempt 1: Estimating instance-specific parameters independently. Recall that our target is to achieve an instance-specific error in (1), which further depends on three instance-specific values, $\varepsilon^{\text{user}}_{\min}(D)$, $\varepsilon^{\text{rec}}_{\min}(D)$ and $m_{\max}(D)$. At first glance, the problem seems decomposable: we can estimate these parameters separately by applying the PDP private estimator from prior work [4]. More precisely, at the record-level, we first privately estimate $\varepsilon^{\text{rec}}_{\min}(D)$ using the existing record-PDP solution, and then use it as a threshold to remove records whose privacy budgets fall below it. Then, similarly, at the user-level, we estimate $\varepsilon^{\text{user}}_{\min}(D)$ and remove users (together with all their records) whose privacy budgets fall below the user-level threshold. After this step, since all users have privacy budgets at least $\tilde{\varepsilon}^{\text{user}}_{\min}(D)$, we can further estimate $m_{\max}(D)$ using a record-DP maximum estimator, as in [9] and [13], with the uniform privacy budget $\tilde{\varepsilon}^{\text{user}}_{\min}(D)$. After that, we cap the number of each user's records at the estimated maximum $\widetilde{m}_{\max}(D)$. Finally, after these three rounds of truncation, we apply a record-DP mechanism with a uniform privacy budget $\varepsilon' = \min\left\{\tilde{\varepsilon}^{\text{rec}}_{\min}(D),\ \frac{\tilde{\varepsilon}^{\text{user}}_{\min}(D)}{\widetilde{m}_{\max}(D)}\right\}$ to all records in the resulting dataset. When these parameters are estimated with sufficient accuracy, we nearly attain the error bound in $\text{Err}_{\mathcal{M}}(D, \Omega(\varepsilon'), \beta)$. The quantity ε' is still smaller than our target $\varepsilon_{\min}(D)$ because it estimates $\varepsilon^{\text{rec}}_{\min}(D)$ and $\varepsilon^{\text{user}}_{\min}(D)$ separately, whereas $\varepsilon_{\min}(D)$ accounts for them jointly. However, ε' is instance-specific and can be considerably larger than the global bound $\check{\varepsilon}$ as well.

From a naive perspective, this approach appears to satisfy Hybrid-PDP, since we estimate both the record- and user-level privacy parameters under privacy constraints separately. However, it ignores the privacy dependency between users and records arising from the fact that each user may contribute multiple records. Consequently, when we estimate $\varepsilon^{\text{rec}}_{\min}(D)$, we consume privacy budget not only at the record-level but also at the user-level, which can be accounted for through group privacy (Lemma 1). For example, suppose a user u with $\Phi_{\text{user}}(u) = 0.2$ has three records, each with privacy budget 0.1. Then record-level estimation exhausting the record-level budget causes a total privacy loss of $0.1 + 0.1 + 0.1 = 0.3 > 0.2$ for u, which violates the user-level privacy requirement.

Attempt 2: A record-level parameters capping-based estimation method. Motivated by the failure of Attempt 1 — in which estimating the minimal record-level parameter in (2) compromises user-level privacy, we refine the procedure by swapping the order of estimation and adding a capping step. More precisely, we

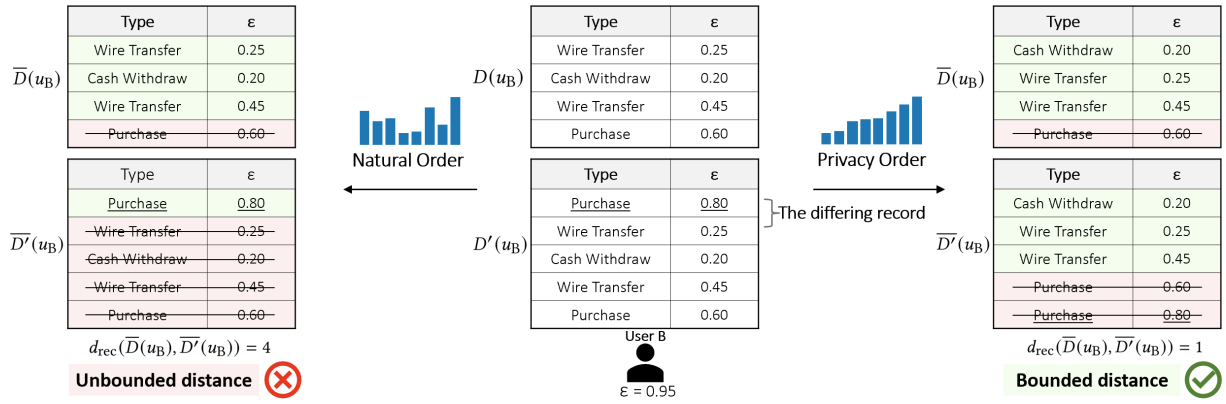


Figure 2: An example illustrating the difference between natural and privacy order-based prefix-truncation schemes.

first privately estimate the user-relevant parameters $\varepsilon_{\min}^{\text{user}}(D)$ and $m_{\max}(D)$, and then truncate users and their records according to the rules in Attempt 1. Then, we further cap each of the record-level privacy budgets by $\tilde{\varepsilon}_{\min}^{\text{user}}(D)/\tilde{m}_{\max}(D)$. After that, we estimate the minimal record-level privacy budget over the clipped dataset. Finally, we run a record-DP mechanism on the resulting dataset with that uniform record-level privacy budget.

This idea successfully bounds user-level privacy leakage during estimation of the minimal record-level privacy budget. Since each user contributes at most $\tilde{m}_{\max}(D)$ records after truncation and each record consumes at most $\tilde{\varepsilon}_{\min}^{\text{user}}(D)/\tilde{m}_{\max}(D)$ of privacy budget, the total privacy loss for any retained user is at most $\tilde{\varepsilon}_{\min}^{\text{user}}(D)$, thereby controlling user-level privacy loss during estimation of the minimal record-level privacy budget. However, this overlooks the fact that privately estimating user-level parameters such as $\varepsilon_{\min}^{\text{user}}(D)$ and $m_{\max}(D)$ also consumes the record-level privacy budget. For example, if each user has a single record, then the existence of a record implies the existence of the corresponding user. Thus, deleting that record may induce a change in $\varepsilon_{\min}^{\text{user}}(D)$. Moreover, $m_{\max}(D)$ depends on per-user record counts in the dataset D. In the extreme case where the dataset contains only one user, inserting or deleting a record necessarily changes $m_{\max}(D)$. Therefore, privately estimating $\varepsilon_{\min}^{\text{user}}(D)$ or $m_{\max}(D)$ must be accounted for within the record-level privacy budget.

Discussion. These attempts reveal a *bidirectional* privacy dependency between the record- and user-levels. Concretely, estimating record-level parameters incurs additional user-level loss via group-privacy, while estimating user-level parameters can still consume record-level budget since record-neighboring changes may alter the user set. Therefore, without additional preprocessing for the datasets, any sequential estimation pipeline that starts at either level tends to under-account for the other and fail to satisfy Hybrid-PDP.

5.2 Prefix-truncation: Dealing with Bidirectional Privacy Dependency

Motivated by the bidirectional privacy dependency, in this section we propose a per-user truncation-based preprocessing step designed to ensure that record-level privacy implies user-level privacy. For each user u, we truncate the records $D(u)$ until the cumulative record-level privacy budget of the retained records is at most $\Phi_{\text{user}}(u)$. After this preprocessing, the record-level privacy guarantee dominates the user-level guarantee via group privacy (Lemma 1). Therefore, the problem reduces to a record-PDP problem, allowing us to leverage existing PDP methods [4] to realize Hybrid-PDP. Here, we first assume an ordering over records. Then, for each user u, we retain only the longest prefix (under this order) whose cumulative record-level privacy budget does not exceed $\Phi_{\text{user}}(u)$. We refer to this idea as *prefix-truncation*.

To perform prefix-truncation, the most straightforward idea is to assume there is a natural order on records (e.g., lexicographic). However, performing truncation according to such an ordering can violate the desired privacy guarantees. The issue is that inserting or deleting a single record for a user can change the positions of many of that user's other records in the ordering, thereby altering which records are retained. As a result, a small change can cascade into multiple changes in the truncated dataset, causing record-level neighbors to become far apart after truncation. For example, consider a user with privacy budget 1 and 100 records, where each record has a privacy budget of 0.01. After prefix-truncation, all records are retained. Now we construct another record-neighboring dataset by inserting a record for the user that has privacy budget 1 and precedes all existing records in the global order. Then this neighboring dataset retains only one record after truncation, leading to an unbounded record-level distance between the original and neighboring truncated datasets.

To avoid the distance blow-up caused by prefix-truncation under an arbitrary predefined order, we instead introduce a *privacy-based ordering*. Specifically, we sort the records in $\mathcal{R}$ in nondecreasing order of $\Phi_{\text{rec}}(r)$, breaking ties according to a fixed global natural order. This ordering is therefore fixed and fully data-independent. In the following, we show that prefix-truncation under this ordering guarantees that any two record-level neighboring datasets remain within a bounded distance after truncation.

Lemma 8 (Record-level Stability of privacy-based order truncation). *For any record-neighboring datasets $D \sim_r D'$, let $\overline{D}$ and $\overline{D}'$ denote the datasets obtained by applying privacy-based*

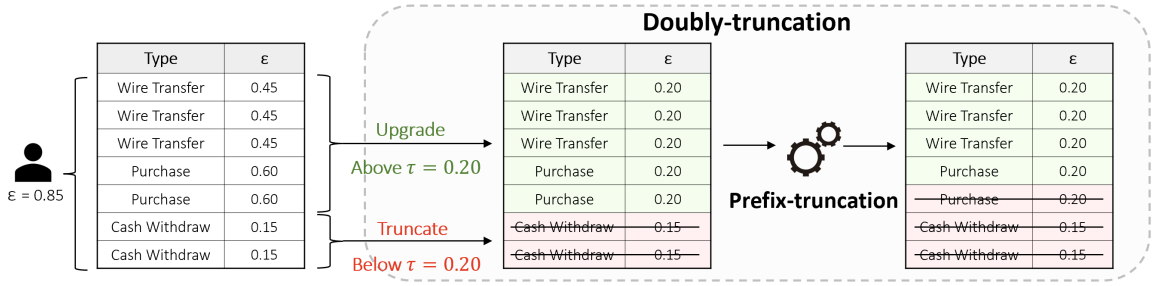


Figure 3: Illustration of the doubly-truncation workflow for a user with $\tau = 0.20$.

ordering prefix-truncation to D and D', respectively. Then the record-level distance $d_{\text{rec}}(\overline{D}, \overline{D}') \leq 2$.

Due to space restrictions, we defer all proofs of the lemmas and theorems to Appendix A. Intuitively, Lemma 8 holds because inserting or deleting any record can allow at most one additional record to be included during prefix-truncation, as all subsequent records have no smaller privacy budgets than the deleted one. Records with lower privacy budgets are not affected, since they appear earlier in the ordering and thus remain within the retained prefix. Specifically, Figure 2 illustrates that, for the neighboring dataset of User B in Figure 1, natural-order truncation can result in an unbounded distance, whereas the privacy order guarantees that, after truncation, the record-level distance is constrained to be at most 2. As a result, such single changes do not trigger cascading effects that can occur under the natural ordering. In the example discussed earlier, the new record with a privacy budget of 1 cannot precede any of those with a privacy budget of 0.01. Therefore, adding it will not cause any of the existing records to be additionally truncated.

Moreover, after prefix-truncation, it is straightforward to show that record-PDP dominates user-PDP via group privacy (Lemma 1).

Lemma 9 (Dominance of record-level privacy over user-level privacy after prefix-truncation). *Let $\overline{D}$ denote the dataset obtained by prefix-truncation. Then, for any user u, $\sum_{r \in \overline{D}(u)} \Phi_{\text{rec}}(r) \leq \Phi_{\text{user}}(u)$.*

Based on the above two properties, we design a Hybrid-PDP mechanism as follows. We first apply prefix-truncation to obtain the retained dataset $\overline{D}$, and then invoke an existing record-PDP mechanism, such as [4], on $\overline{D}$ with the modified privacy budget $\Phi'_{\text{rec}}(r) := \Phi_{\text{rec}}(r)/2$ to account for the fact that prefix-truncation may increase the record-level distance by a factor of 2. It is easy to show that this algorithm satisfies Hybrid-PDP: Lemma 8 establishes record-level privacy, while Lemma 9 implies user-level privacy.

Although this mechanism successfully achieves Hybrid-PDP, its error is unbounded. Below, we show that, in some cases, it can even incur a much larger error than the naive solution that uses a predefined privacy budget. We illustrate this with an example. The dataset contains only a single user u with privacy budget 1 and 1,000,000 records, where each record has a privacy budget of 1.1 and a value of 100. We further suppose that the global record-level budget is $\check{\varepsilon} = 0.001$, that the upper bound on the value of each record is 1000, and that we want to perform a summation task over the dataset. In the prefix-truncation-based mechanism, it is obvious that no records are retained after truncation because the user's privacy budget is smaller than the privacy budget of each record it owns. Therefore, the subsequent record-PDP mechanism operates on an empty dataset, so the error induced by the prefix-truncation can be as large as 100,000,000. For our straw-man framework, we can easily compute the total error of 1,000,000, which is obviously smaller than that of the prefix-truncation-based mechanism. Thus, the overall error of the prefix-truncation-based mechanism cannot be bounded in terms of the instance-specific target in (1).

5.3 Doubly-truncation: Adapting Record-level Privacy to User-level Privacy

The issue with prefix-truncation arises when the user-level privacy constraint is relatively stricter than the record-level constraint. In such cases, the truncation process may delete a large proportion of records to satisfy user-level privacy requirements, or even drop all records in the worst case. One key observation is that existing general PDP methods [4] ultimately reduce the problem to record-DP, by applying record-DP techniques under a uniform privacy budget τ shared by all records. That said, even for records with larger privacy budgets, the privacy loss in the downstream record-DP mechanism is bounded by τ. Consequently, maintaining heterogeneous record-level budgets during the prefix-truncation step becomes redundant and wastes the user's budget for all records with $\Phi_{\text{rec}}(r) > \tau$. Another insight is that we can upgrade the privacy protection of records to a stricter level, i.e., set all $\Phi_{\text{rec}}(r) > \tau$ to τ, which allows more records to be retained during prefix-truncation.

Inspired by the above insights, we propose a *doubly-truncation* strategy (Algorithm 1), in which we first truncate some records under stricter privacy protection, and then upgrade the privacy of the remaining records before applying prefix-truncation. Since all retained records after the first truncation are upgraded to the same privacy budget, using the natural order also satisfies the privacy-based ordering requirement in this case. We design the following algorithm.

Precisely, for a given threshold τ, we drop all records with privacy budgets below τ, which ensures that all records have privacy budgets above τ, allowing us to invoke record-DP algorithms with privacy budget τ to guarantee record-level privacy. Then, we upgrade the remaining records' privacy budgets to τ, allowing more records to be retained during the downstream per-user prefix-truncation. After that, we perform prefix-truncation for each user to ensure

Algorithm 1: Doubly-truncation

Input: Dataset D, privacy functions Φ_{rec}, Φ_{user}, and a threshold τ

Output: The retained dataset $\overline{\overline{D}}$

foreach $u \in D$ **do**
  $D_\tau(u) \leftarrow \{r \in D(u) \,:\, \Phi_{\text{rec}}(r) \geq \tau\}$
  $\kappa(u) \leftarrow \min\left(\left\lfloor \frac{\Phi_{\text{user}}(u)}{\tau} \right\rfloor, \, |D_\tau(u)|\right)$
  $\overline{\overline{D}}(u) \leftarrow$ first $\kappa(u)$ records of $D_\tau(u)$ under the natural order
end
$\overline{\overline{D}} \leftarrow \bigcup_{u \in D} \overline{\overline{D}}(u)$
return $\overline{\overline{D}}$

Algorithm 2: Threshold Selection

Input: Dataset D, privacy functions Φ_{rec}, Φ_{user}, and failure rate β

Output: Selected threshold τ_i

$\ell = \lceil \log(\hat{\varepsilon}/\check{\varepsilon}) \rceil + 1$
for $i \leftarrow 1$ **to** ℓ **do**
  $\tau_i \leftarrow 2^{i-1} \cdot \check{\varepsilon}$
  $\overline{\overline{D}}_i \leftarrow \textsc{Doubly-Truncation}\big(D, \Phi_{\text{rec}}, \Phi_{\text{user}}, \tau_i\big)$
  $\widetilde{S}_i \leftarrow \left|\overline{\overline{D}}_i\right| - \frac{3\ell}{\tau_i} \ln(\ell/\beta) + \text{Lap}(\ell/\tau_i)$
end
$i^\star \leftarrow \arg\max_{1 \leq i \leq \ell} \widetilde{S}_i$
return $\tau_{i^\star}$

that satisfying the record-level privacy guarantee also implies satisfaction of the user-level privacy guarantee. Figure 3 illustrates an example of doubly-truncation for a user with $\tau = 0.20$. Moreover, doubly-truncation can be shown to satisfy the same two properties as prefix-truncation:

Lemma 10 (Record-level Stability of doubly-truncation). *For any record-neighboring datasets $D \sim_r D'$, let $\overline{\overline{D}}$ and $\overline{\overline{D'}}$ denote the datasets obtained by applying doubly-truncation to D and D' with the same truncation threshold. Then the record-level distance $d_{\text{rec}}(\overline{\overline{D}}, \overline{\overline{D'}}) \leq 2$.*

Lemma 11 (Dominance of record-level privacy over user-level privacy after doubly-truncation). *Let $\overline{\overline{D}}$ denote the dataset obtained by applying doubly-truncation with threshold τ. Then, for any user u, $\sum_{r \in \overline{\overline{D}}(u)} \tau \leq \Phi_{\text{user}}(u)$.*

With these two desired properties, we can design a Hybrid-PDP mechanism that first applies doubly-truncation with threshold τ and then invokes a record-DP mechanism with privacy budget $\tau/2$. This approach induces an error trade-off governed by the choice of τ. Intuitively, an overly large threshold results in more records being dropped, thereby reducing data utility, whereas an overly small threshold introduces excessive DP error in the downstream record-DP mechanisms. One appropriate choice is to set the threshold to $\varepsilon_{\min}(D)$ as in (2). With this threshold, all records are retained after the doubly-truncation step, and the DP error introduced by the threshold is controlled by our desired instance-specific quantity. Nevertheless, as established in Section 5.1, any attempt to explicitly estimate $\varepsilon_{\min}(D)$ inevitably fails due to the bidirectional privacy dependency between the record- and user-levels.

5.4 Threshold Selection in Doubly-truncation

Recall that our goal is to determine a value of τ for the doubly-truncation process that is as large as possible while still retaining as many records as possible. To reduce the search space of τ, we consider $\ell = \lceil \log(\hat{\varepsilon}/\check{\varepsilon}) \rceil + 1$ candidate thresholds.[2] More precisely, we set the i-th threshold as $\tau_i = 2^{i-1} \cdot \check{\varepsilon}$. For these thresholds, we use a score function to measure their utilities, and we want to privately select a τ_i with the highest score.

[2]All logarithms denoted by log are base 2 throughout this paper.

Here, our score function consists of two components: the first component is the number of records retained after applying doubly-truncation under the current threshold. The second component is a penalty term that is inversely proportional to the threshold. This term prevents selecting a threshold τ that is too small, i.e., $\tau \ll \varepsilon_{\min}(D)$ as defined in (2), since such a choice would result in adding excessive DP error to the downstream record-DP mechanism. The formal definition of the score function is given by:

$$S_i := \left|\overline{\overline{D}}_i\right| - \frac{3\ell}{\tau_i} \ln(\ell/\beta)$$

where $\overline{\overline{D}}_i$ denotes the retained dataset after doubly-truncation, while $-\frac{3\ell}{\tau_i} \ln(\ell/\beta)$ is the penalty term. With high probability, as we will show later, this term prevents the privatized score function from selecting an overly small threshold.

We now discuss how to privatize each S_i. First, it is easy to show that upgrading the privacy and applying prefix-truncation do not enlarge the record-level sensitivity of S_i:

Lemma 12. *For any threshold τ_i, the corresponding score function S_i changes by at most 1 between any pair of record-neighboring datasets $D \sim_r D'$.*

Lemma 12 holds because, in the first step of the doubly-truncation procedure, i.e., truncating records with privacy budgets below τ_i, inserting or deleting any record only affects whether that record is included in the intermediate retained dataset $\overline{D}$. In the second step, i.e., prefix-truncation, by Lemma 8, we know that inserting any record can cause zero or one other record to be dropped. Combining these observations for the two steps, we conclude that, for the entire doubly-truncation procedure, a single insertion or deletion in the original dataset D can change the size of the final retained dataset $\overline{\overline{D}}_i$ by at most 1.

To privatize all ℓ scores, the idea is to add Laplace noise with a scale of ℓ/τ_i. Then, by Lemma 12, estimating each score function preserves Φ_{rec}/ℓ-record-PDP. Moreover, Lemma 11 ensures that any downstream mechanism satisfying record-level privacy also satisfies user-level privacy. Therefore, estimating each score satisfies $(\Phi_{\text{rec}}/\ell, \Phi_{\text{user}}/\ell)$-Hybrid-PDP. By sequential composition, the entire estimation process, which has ℓ scores, complies with the desired privacy guarantees of $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP.

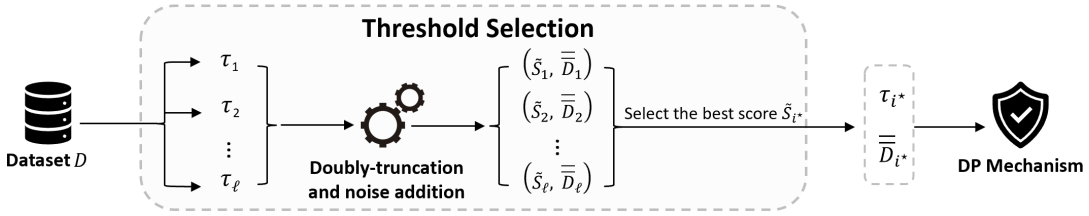


Figure 4: Illustration of the workflow of our proposed Hybrid-PDP framework.

To conclude, the privatized score function can be set as

$$\widetilde{S}_i := \left|\overline{\overline{D}}_i\right| - \frac{3\ell}{\tau_i}\ln(\ell/\beta) + \mathrm{Lap}(\ell/\tau_i).$$

The detailed algorithm for privately estimating the best threshold is given in Algorithm 2.

Theorem 5.1. *For any given dataset D and any given privacy functions Φ_{rec} and Φ_{user}, Algorithm 2 satisfies $(\Phi_{\mathrm{rec}}, \Phi_{\mathrm{user}})$-Hybrid-PDP and, with probability at least $1-\beta$, outputs $\tau_{i^\star}$ such that the following hold:*

(i) $\tau_{i^\star} \geq \frac{1}{2}\,\varepsilon_{\min}(D)$.

(ii) $\left|\,|D| - |\overline{\overline{D}}_{i^\star}|\right| = O\left(\frac{1}{\varepsilon_{\min}(D)}\log(\hat{\varepsilon}/\check{\varepsilon})\log\frac{\log(\hat{\varepsilon}/\check{\varepsilon})}{\beta}\right)$, *where $\overline{\overline{D}}_{i^\star}$ is obtained by applying doubly-truncation to D with threshold $\tau_{i^\star}$.*

In conclusion, using Algorithm 2, we can estimate an appropriate threshold that is neither too small, i.e., larger than $\frac{1}{2}\,\varepsilon_{\min}(D)$ to ensure that only instance-specific DP noise is introduced in the subsequent record-DP mechanism, nor too large, so that it excludes at most $\tilde{O}(1/\varepsilon_{\min}(D))$ records from the original dataset. Thus, we set the selected threshold as a uniform privacy budget over the retained dataset, and then invoke a downstream record-DP mechanism with the selected $\tau_{i^\star}$.

5.5 Overall Mechanism

After showing how to select an appropriate $\tau_{i^\star}$ under Hybrid-PDP, we are now ready to present our final Hybrid-PDP framework. Given any existing record-DP mechanism, our framework can extend it to Hybrid-PDP. In this framework, in Stage 1, we run Algorithm 2 with $\Phi_{\mathrm{rec}}/2$ and $\Phi_{\mathrm{user}}/2$ to obtain a threshold $\tau_{i^\star}$ with the guarantee claimed in Theorem 5.1. [3] Then, in Stage 2, we perform doubly-truncation with the threshold $\tau_{i^\star}$ over D with Algorithm 1 to obtain the truncated dataset $\overline{\overline{D}}_{i^\star}$. Afterward, we apply the given record-DP mechanism with privacy budget $\tau_{i^\star}/4$. The factor $1/4$ arises from two sources. First, half of the overall privacy budget is spent in Stage 1 to estimate $\tau_{i^\star}$. Therefore, in Stage 2, the remaining privacy budget available to each record and each user is reduced by a factor of 2, and the effective truncation threshold becomes $\tau_{i^\star}/2$. Second, doubly-truncation can enlarge the record-level distance by a factor of 2. To compensate for this enlargement, the privacy budget of the record-DP mechanism is reduced by another factor of 2.[4] The detailed algorithm is presented in Algorithm 3, and Figure 4 illustrates its workflow.

Algorithm 3: Instance-Specific Hybrid-PDP Mechanism

Input: A dataset D, arbitrary record-DP mechanism $\mathcal{M}$, privacy budget functions $\Phi_{\mathrm{rec}}, \Phi_{\mathrm{user}}$, and failure rate β

Output: $\widetilde{Q}(D)$ under $(\Phi_{\mathrm{rec}}, \Phi_{\mathrm{user}})$-Hybrid-PDP

```
// Stage 1: Threshold selection:
```

$\tau^{\mathrm{half}}_{i^\star} \leftarrow \textsc{ThresholdSelection}(D, \Phi_{\mathrm{rec}}/2, \Phi_{\mathrm{user}}/2, \beta/2)$

$\tau_{i^\star} \leftarrow 2\tau^{\mathrm{half}}_{i^\star}$

```
// Stage 2: Run a record-DP mechanism on the
   truncated dataset:
```

$\overline{\overline{D}}_{i^\star} \leftarrow \textsc{Doubly-Truncation}\big(D, \Phi_{\mathrm{rec}}, \Phi_{\mathrm{user}}, \tau_{i^\star}\big)$

$\widetilde{Q}(D) \leftarrow \mathcal{M}\big(\overline{\overline{D}}_{i^\star}, \tau_{i^\star}/4, \beta/2\big)$

return $\widetilde{Q}(D)$

For efficiency, suppose there are M users and N records. Algorithm 2 runs in $O(\log(\hat{\varepsilon}/\check{\varepsilon})\,(M+N))$ time, because it performs $\lceil\log(\hat{\varepsilon}/\check{\varepsilon})\rceil$ rounds, each involving a doubly-truncation that takes $O(M+N)$ time. For privacy, by Theorem 5.1, τ_{i^*} computed in Stage 1 of Algorithm 3 satisfies $(\Phi_{\mathrm{rec}}/2, \Phi_{\mathrm{user}}/2)$-Hybrid-PDP. Moreover, given τ^*, Lemma 10 and Lemma 11 imply that Stage 2 satisfies $(\Phi_{\mathrm{rec}}/2, \Phi_{\mathrm{user}}/2)$-Hybrid-PDP. By sequential composition, Algorithm 3 satisfies $(\Phi_{\mathrm{rec}}, \Phi_{\mathrm{user}})$-Hybrid-PDP. For utility, we establish the following guarantee:

Theorem 5.2. *For any privacy functions Φ_{user} and Φ_{rec}, any $\beta \in (0,1)$, and any record-DP mechanism $\mathcal{M}$ for query Q, Algorithm 3 satisfies $(\Phi_{\mathrm{rec}}, \Phi_{\mathrm{user}})$-Hybrid-PDP and, with probability at least $1-\beta$, returns $\widetilde{Q}(D)$ such that the following holds:*

$$\left|\widetilde{Q}(D) - Q(D)\right| \leq \underbrace{\left|Q(D) - Q(D')\right|}_{I} + \underbrace{\mathrm{Err}_{\mathcal{M}}\big(D', \varepsilon_{\min}(D)/8, \beta/2\big)}_{II},$$

where D' is obtained by removing

$$O\left(\frac{1}{\varepsilon_{\min}(D)}\log(\hat{\varepsilon}/\check{\varepsilon})\log\frac{\log(\hat{\varepsilon}/\check{\varepsilon})}{\beta}\right)$$

records from D.

[3] Strictly speaking, Algorithm 2 first returns $\tau^{\mathrm{half}}_{i^\star}$, which satisfies Theorem 5.1 with respect to the halved privacy functions. By setting $\tau_{i^\star} = 2\tau^{\mathrm{half}}_{i^\star}$, the guarantees of Theorem 5.1 translate back to the original privacy functions with the same form.

[4] This distance metric used in Lemma 10 is the insert-one/delete-one distance. If we consider the change-one distance, the distance remains 1. Therefore, if the given record-DP mechanism works under both the insert-one/delete-one and change-one settings, then we can use $\tau_{i^\star}/2$ as the privacy budget.

As shown in the above theorem, the error consists of two components. Component I arises from excluding $\tilde{O}(1/\varepsilon_{\min}(D))$ records during the doubly-truncation step. Under the DP setting, it has been shown [4] that any ε-DP mechanism at least incurs an error of $|Q(D) - Q(D')|$, where D' is an arbitrary dataset obtained from D by removing $\tilde{O}(1/\varepsilon_{\min}(D))$ records. Therefore, Component I is at most $\mathrm{Err}_{\mathcal{M}}(D, \Omega(\varepsilon_{\min}(D)), \beta)$ by a logarithmic factor.

For Component II, which arises from the intrinsic DP error of the downstream record-DP mechanism, since DP error is typically inversely proportional to the privacy budget, this term is $O(\mathrm{Err}_{\mathcal{M}}(D', \Omega(\varepsilon_{\min}(D)), \beta))$. Moreover, since D and D' differ by only $\tilde{O}(1/\varepsilon_{\min}(D))$ number of records, in most cases,

$$O\big(\mathrm{Err}_{\mathcal{M}}(D', \Omega(\varepsilon_{\min}(D)), \beta)\big) \approx O\big(\mathrm{Err}_{\mathcal{M}}(D, \Omega(\varepsilon_{\min}(D)), \beta)\big).$$

Overall, the above error is bounded by $O(\mathrm{Err}_{\mathcal{M}}(D, \Omega(\varepsilon_{\min}(D)), \beta))$. Below, we will show the error guarantees for several common tasks under our mechanism.

Sum estimation. For sum estimation, we first denote $F(r)$ as the value of record r, which is defined over the range $[1, 2, \ldots, B]$. Under this notation, the most straightforward solution in record-DP is the Laplace mechanism, which achieves an error bound of $O(B/\varepsilon)$. Furthermore, the state-of-the-art (SOTA) summation estimator from [9] achieves an error bound of $\tilde{O}\big(\max_{r\in D} F(r)/\varepsilon\big)$. This instance-specific error is more desirable since $\max_{r\in D} F(r) \ll B$ for most datasets. In our framework, Component I in Theorem 5.2 incurs an error of $\tilde{O}(\max_{r\in D} F(r)\ /\varepsilon_{\min}(D))$. For Component II, invoking [9] as the downstream record-DP mechanism with a privacy budget over D' yields an error of $\tilde{O}(\max_{r\in D} F(r)\ /\varepsilon_{\min}(D'))$. Since this instance-specific error is supermodular, it can be further bounded by $\tilde{O}(\max_{r\in D} F(r)\ /\varepsilon_{\min}(D))$. Thus, the overall error is $\tilde{O}(\max_{r\in D} F(r)\ /\varepsilon_{\min}(D))$, which aligns with our target.

Distinct count. For distinct count, Component I is $\tilde{O}(1/\varepsilon_{\min}(D))$. Meanwhile, Component II is $O(1/\varepsilon_{\min}(D'))$. Since the error of distinct count is supermodular, it can also be bounded by $\tilde{O}(1/\varepsilon_{\min}(D))$. Combining these yields a total error of $\tilde{O}(1/\varepsilon_{\min}(D))$, which is at most a doubly-logarithmic factor larger than the record-DP error.

Maximum estimation. For maximum estimation, we use *rank error* as the metric, which measures the true rank of the estimated maximum. In record-DP, we can invoke [9] to obtain a maximum with rank error $\tilde{O}(1/\varepsilon_{\min}(D))$. In our mechanism, using [9] as the base DP mechanism, Component I yields a rank error of $\tilde{O}(1/\varepsilon_{\min}(D))$, while Component II also contributes an error of $\tilde{O}(1/\varepsilon_{\min}(D))$. Therefore, the overall rank error remains $\tilde{O}(1/\varepsilon_{\min}(D))$, as desired.

6 EXPERIMENTS

In this section, we present empirical results for our proposed Hybrid-PDP framework under a variety of privacy specifications and across both real-world and synthetic settings. We evaluate the following three fundamental queries. For *Count*, we incorporate the classic Laplace mechanism (Lemma 4) into our framework. For *Sum*, we integrate the SOTA Laplace-based, instance-specific standard DP *Sum* mechanism [9]. For *Max*, we incorporate the Laplace-based exponential mechanism introduced in [9].

For all queries, we compare our framework against two baselines: (i) a naive approach that applies global privacy bounds to both user-level and record-level privacy budgets; and (ii) the prefix-truncation method proposed in Section 5.2, where downstream queries are answered using the SOTA record-level PDP methods from [4]. For evaluation metrics, we measure errors using relative error for *Count* and *Sum*, and use the more meaningful relative rank error for *Max*.

6.1 Setup

Datasets. We consider a variety of datasets from different real-world settings, including banking transactions (Czech Bank [20]), salary payments (CK Pay [5], NY Salary [33]), retail statistics (Online Retail [19]), web user behavior data (VLE [22]), and the standard benchmark databases (TPCH Order and TPCH Lineitem [37]). In total, we use five real-world datasets, two TPC-H benchmark datasets, and a suite of generated synthetic datasets, covering diverse application scenarios and varying dataset sizes and distributions, as shown in Table 1. Common preprocessing practices, such as excluding abnormal values are applied to all datasets. For more information on dataset-specific preprocessing techniques and further details of the synthetic datasets, please refer to Appendix B.1.

Privacy Budgets. We adopt log-scale randomization for both user- and record-level privacy budgets, as it yields greater heterogeneity and better captures diverse privacy requirements. This setting is well accepted in the DP research community [7, 41]. For user-level privacy budgets, all real-world datasets share the same privacy specification. Specifically, for each user or user group, defined by certain attributes in the original databases, the privacy budget is independently sampled according to $\Phi_{\text{user}}(u) \propto 2^{\mathcal{N}(0,\sigma)}$. For record-level privacy budgets, we consider three different privacy budget functions to demonstrate the generalizability of our framework. For the CK Pay, NY Salary, and VLE datasets, we adopt the log-normal function $\Phi_{\text{rec}}(r) \propto 2^{\mathcal{N}(0,\sigma)}$. For the Online Retail and Czech Bank datasets, we use the linear function $\Phi_{\text{rec}}(r) \propto 2^{\mathcal{N}(0,\sigma)} \cdot F(r)$. For the TPCH Order and TPCH Lineitem datasets, we use the square-root function $\Phi_{\text{rec}}(r) \propto 2^{\mathcal{N}(0,\sigma)} \cdot (F(r))^{0.5}$. In all cases, privacy budgets are independently sampled for each record or record group, as defined by specific attributes in the original databases. For further details, please refer to Appendix B.2.

Other Parameters. All the empirical results are obtained using a Linux server equipped with dual AMD EPYC 9555 64-Core Processors and 2 TB of system memory. For the global bounds on user- and record-level privacy budgets, we set $\check{\varepsilon}_{\text{user}} = 2^{-10}$, $\check{\varepsilon}_{\text{rec}} = 2^{-20}$, $\hat{\varepsilon}_{\text{user}} = \hat{\varepsilon}_{\text{rec}} = 2^4$, and $\hat{m} = 2^{16}$. We set the failure rate to $\beta = 10\%$ across all experiments. For each setting, we repeat the experiment 50 times, drop the top and bottom 10 results, and report the average over the remaining 30 runs for both relative error (RE) and running time. This 40% trimming is more conservative than the failure rate and helps ensure reliability.

6.2 Results

6.2.1 Utility and Efficiency. Table 2 presents the empirical results of our methods on the three investigated queries across seven datasets, compared with baseline methods, demonstrating the superior data utility of our framework. It is evident that the naive global-bound

Dataset	# of users	# of records	$\text{med}_{u \in D}(m(D,u))$	$\max_{u \in D}(m(D,u))$	$\text{med}_{r \in D}(F(r))$	$\max_{r \in D}(F(r))$
CK Pay	$\sim 5.3 \times 10^4$	$\sim 2.3 \times 10^5$	2	14	$\sim 1.6 \times 10^4$	$\sim 2.4 \times 10^5$
NY Salary	$\sim 2.1 \times 10^5$	$\sim 8.9 \times 10^5$	4	59	$\sim 7.5 \times 10^4$	$\sim 2.0 \times 10^6$
VLE	$\sim 2.6 \times 10^4$	$\sim 1.1 \times 10^7$	270	6389	2	$\sim 7.0 \times 10^3$
Online Retail	$\sim 2.6 \times 10^4$	$\sim 5.4 \times 10^5$	10	1114	10	$\sim 1.7 \times 10^5$
Czech Bank	$\sim 4.5 \times 10^3$	$\sim 1.1 \times 10^6$	12	675	12	$\sim 8.7 \times 10^4$
TPCH Order	$\sim 1.0 \times 10^5$	$\sim 1.5 \times 10^6$	14	41	$\sim 1.4 \times 10^5$	$\sim 5.6 \times 10^5$
TPCH Lineitem	$\sim 1.0 \times 10^4$	$\sim 6.0 \times 10^6$	600	694	$\sim 9.5 \times 10^5$	$\sim 5.2 \times 10^6$
Synthetic Datasets	$\sim 7.8 \times 10^3 \sim 6.3 \times 10^5$	$\sim 1.0 \times 10^6$	$1 \sim 128$	$9 \sim 171$	$\sim 1.3 \times 10^6$	$\sim 5.6 \times 10^6$

Table 1: Dataset statistics. Here, med denotes the median, and max denotes the maximum value. For the synthetic datasets, we report ranges over all datasets used, since they have different user-record ownership distributions.

Dataset			CK Pay	NY Salary	VLE	Online Retail	Czech Bank	TPCH Order	TPCH Lineitem
Privacy Specification			$\Phi_{\text{rec}}(r) \propto 2^{N(0,\sigma)}$			$\Phi_{\text{rec}}(r) \propto F(r) \cdot 2^{N(0,\sigma)}$		$\Phi_{\text{rec}}(r) \propto \sqrt{F(r)} \cdot 2^{N(0,\sigma)}$	
Count	Naive	RE (%)	>100	>100	>100	>100	>100	>100	>100
		Time (ms)	0.32	1.99	55.9	1.03	3.66	3.52	27.3
	Prefix	RE (%)	0.1303	0.1138	91.1	48.2	62.1	17.84	83.5
		Time (s)	0.35	2.01	9.28	0.56	2.17	6.70	10.6
	Framework	RE (%)	0.00302	0.1116	0.1387	5.40	16.35	0.957	0.881
		Time (s)	0.61	2.67	24.9	1.02	3.01	5.31	20.1
Sum	Naive	RE (%)	100	100	100	100	100	100	100
		Time (ms)	17.1	62.8	819	40.3	78.8	108	457
	Prefix	RE (%)	0.256	0.233	93.6	79.0	98.9	25.7	99.2
		Time (s)	0.46	2.49	10.3	0.67	1.92	8.48	13.1
	Framework	RE (%)	0.1813	0.1260	3.16	14.31	2.11	0.0378	0.970
		Time (s)	0.71	3.14	31.5	1.20	4.19	8.69	42.1
Max	Naive	RE (%)	>100	>100	>100	>100	>100	>100	>100
		Time (s)	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
	Prefix	RE (%)	0.472	0.1243	0.255	12.43	67.5	1.122	74.1
		Time (s)	0.38	2.16	9.32	0.59	2.19	6.79	11.0
	Framework	RE (%)	0.469	0.1258	1.352	1.644	1.707	0.1447	0.594
		Time (s)	0.66	2.88	24.0	1.05	3.19	5.93	22.9

Table 2: Comparison of different mechanisms under Hybrid-PDP for count, sum, and max. RE (%) denotes relative error.

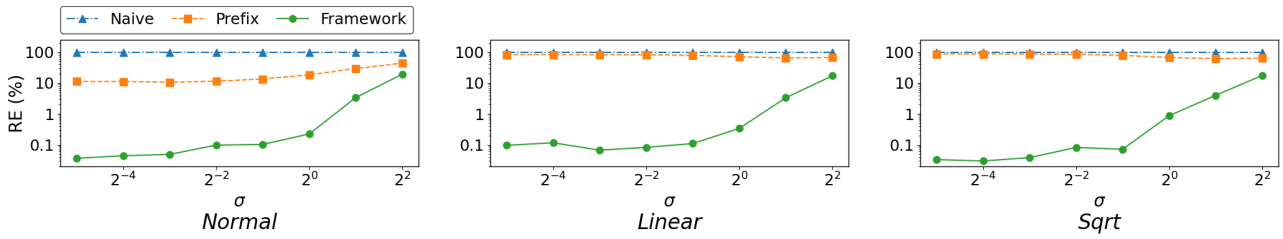


Figure 5: Comparison of RE (%) for *Sum* under Hybrid-PDP across three privacy specifications and varying privacy budget skewness σ. Both axes are in log scale.

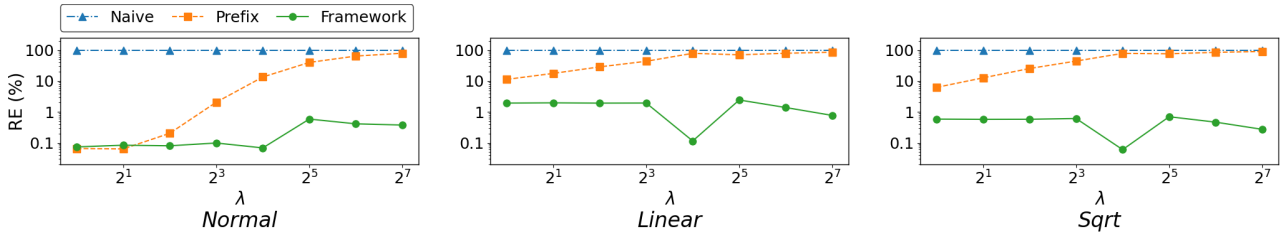


Figure 6: Comparison of RE (%) for *Sum* under Hybrid-PDP across three privacy specifications and varying user-record ownership distributions. Both axes are in log scale.

baseline offers no utility, with errors consistently at 100% or higher. This is because the global bound is overly conservative compared with the actual records or users with the most stringent privacy requirements in each dataset, and using it therefore leads to poor

performance. More specifically, for the *Max* query, such small privacy budgets can even violate the applicability conditions of the base exponential mechanism, which requires $|D| > \tilde{O}(1/\varepsilon)$.

The prefix-truncation method can sometimes achieve acceptable utility, especially on CK Pay and NY Salary. This is expected because each user owns relatively fewer records in these two datasets, as shown in Table 1. However, for datasets in which each user has a large average number of records, such as VLE, prefix truncation often yields REs above 90%. This shows that prefix-truncation alone is often insufficient to achieve good utility.

In contrast, our framework with doubly-truncation consistently outperforms prefix-truncation across almost all datasets and queries under diverse privacy settings, while maintaining runtime efficiency. It achieves up to a 600× utility improvement with only up to a 3× runtime overhead. In many cases, especially for the more complex *Sum* and *Max* queries, these utility gains are essential for achieving high data utility. For example, in the *Sum* query on the VLE dataset, the prefix-truncation method yields an RE of nearly 100%, whereas our framework reduces it to only 3.16%. Similarly, in *Max* on the Czech Bank dataset, the error is reduced from 67.5% to 1.707%. Collectively, these findings provide strong empirical evidence for the effectiveness of our doubly-truncation framework in achieving Hybrid-PDP efficiently while preserving high data utility.

6.2.2 Privacy Budget Skewness. To demonstrate the impact of privacy budget skewness, we report empirical results for *Sum* on the synthetic datasets, under three privacy specifications. More specifically, we fix the user-level privacy budget function as $\Phi_{\text{user}}(u) = 2^2 \cdot 2^{\mathcal{N}(0,\sigma)}$, and consider three record-level privacy budget functions: normal ($\Phi_{\text{rec}}(r) = 2^{-2} \cdot 2^{\mathcal{N}(0,\sigma)}$), linear ($\Phi_{\text{rec}}(r) = 2^{-20} \cdot 2^{\mathcal{N}(0,\sigma)} \cdot F(r)$), and sqrt ($\Phi_{\text{rec}}(r) = 2^{-10} \cdot 2^{\mathcal{N}(0,\sigma)} \cdot (F(r))^{0.5}$). We vary the standard deviation σ from 2^{-5} to 2^2, with larger σ corresponding to greater skewness.

As shown in Figure 5, the utility of our framework consistently outperforms the baselines across all settings, with RE below 20%, even when σ is very large, indicating extremely skewed privacy budget distributions. The prefix-truncation-based method provides acceptable utility, with RE slightly above 20%, only under moderate σ for the normal privacy budget function. In other cases, the prefix-truncation method does not provide significant performance gains compared with the naive baselines, yielding nearly 100% REs. These results highlight the critical role of our proposed doubly-truncation technique in ensuring high data utility under Hybrid-PDP.

6.2.3 User-record Ownership Distribution. To study the impact of the user-record ownership distribution, we report empirical results for *Sum* under different Poisson ownership distributions, $\Pr(m(D,u) = k) = \lambda^k e^{-\lambda}/k!$, varying λ from 2^0 to 2^7, where larger λ indicates a higher expected number of records owned by each user. We also include the three privacy specifications considered in Section 6.2.2, with $\sigma = 2^{-1}$ fixed to represent moderate privacy budget skewness.

Figure 6 shows that our proposed framework achieves the best data utility across all settings, with REs consistently below 3%. Consistent with the results in Section 5.2, the prefix-truncation technique alone is insufficient in many cases. In particular, when users have a relatively larger expected number of records, which means the user-level privacy constraint becomes dominant, the RE of the prefix-truncation increases rapidly, highlighting the necessity of doubly-truncation. This is because the prefix-truncation method discards significantly more records as the number of records per user increases, whereas the doubly-truncation framework mitigates this effect by selecting a good τ to cap each record's budget, allowing more records to be retained, consistent with Section 5.3.

Dataset	CK Pay	Online Retail	Czech Bank	TPCH Order
Framework	0.1813	14.31	2.11	0.0378
$\tau = 2^{-9}$	4.38	18.97	2.12	0.578
$\tau = 2^{-8}$	2.01	14.17	1.114	0.1563
$\tau = 2^{-7}$	1.494	11.94	1.780	0.1153
$\tau = 2^{-6}$	0.726	10.86	14.84	0.0710
$\tau = 2^{-5}$	0.486	12.55	39.6	0.0277
$\tau = 2^{-4}$	0.1658	18.03	61.3	0.0355
$\tau = 2^{-3}$	0.0849	26.9	77.3	0.800
$\tau = 2^{-2}$	47.8	41.2	86.9	15.22
Opt. Ratio (%)	213.5	131.8	189.4	136.5

Table 3: Comparison of RE (%) for *Sum* achieved by our framework versus baselines with predefined τ values on four datasets.

6.2.4 Optimality of τ. Since our method relies on selecting an optimal threshold τ, we conduct an ablation study by manually varying τ over the range 2^{-9} to 2^{-2}. We compare the resulting REs with those obtained by our framework on four datasets for *Sum*, under the same privacy settings as in previous sections.

As shown in Table 3, performance is highly sensitive to the choice of τ and can vary significantly across datasets: an inappropriate value just 2× larger than the optimal can lead to more than a 500× increase in RE. This underscores that selecting τ is critical for data utility, yet choosing an appropriate value remains challenging. Nevertheless, our framework consistently achieves REs within a small constant factor of those obtained with the best manually tuned τ, as indicated by the optimality ratio, defined as Optimality Ratio = RE(Framework)/RE(manually set optimal τ) × 100%. This is expected, since our framework allocates a portion of the privacy budget to privately estimating τ, whereas the baselines are strengthened by manually tuning τ to attain their best utility and are therefore not fully DP. Overall, these results show that our framework can accurately estimate an appropriate τ for each case, thereby achieving high data utility.

7 CONCLUSION

In this work, we propose Hybrid-PDP, a novel differential privacy model that provides personalized privacy guarantees simultaneously at both the record- and user-levels. We develop a general framework that reduces the Hybrid-PDP problem to a record-DP problem, enabling the use of any DP mechanism to achieve Hybrid-PDP protection. Several directions remain open for future work. One is to extend Hybrid-PDP to dynamic settings, where streams of users and records arrive over time. Another interesting direction is to study scenarios in which a record is jointly owned by multiple users [34]. In relational databases, this corresponds to settings involving foreign-key constraints.

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A PROOFS OF THEOREMS AND LEMMAS

A.1 Proof of Lemma 8

Proof. By symmetry of record-neighboring and of $d_{\text{rec}}(\cdot,\cdot)$, it suffices to consider the case where D' is obtained from D by deleting a single record r^*. Let u^* be the user owning r^*.

For every user $u \neq u^*$, the multi-set of records owned by u is identical in D and D'. Hence the privacy-ordered list of records for u, and therefore the retained prefix after prefix-truncation, are identical in $\overline{D}$ and $\overline{D}'$. Thus only the records of u^* can contribute to $d_{\text{rec}}(\overline{D}, \overline{D}')$. Let

$$r_1 \prec r_2 \prec \cdots \prec r_m$$

be the records of u^* in D, listed in privacy-based order. Assume that ties in the privacy-based order are broken by a fixed deterministic rule independent of the dataset. By definition of this order,

$$\Phi_{\text{rec}}(r_1) \leq \Phi_{\text{rec}}(r_2) \leq \cdots \leq \Phi_{\text{rec}}(r_m). \tag{3}$$

Assume that $r^* = r_t$ for some $t \in \{1, \ldots, m\}$, and define

$$C_j := \sum_{i=1}^{j} \Phi_{\text{rec}}(r_i), \qquad C_0 := 0.$$

Let

$$k := \max\big\{j \in \{0, \ldots, m\} : C_j \leq \Phi_{\text{user}}(u^*)\big\}. \tag{4}$$

Then the retained prefix of user u^* in $\overline{D}$ is

$$P := \{r_1, \ldots, r_k\}.$$

If $k = m$, then all records of u^* are retained in $\overline{D}$. After deleting r_t, the retained prefix in $\overline{D}'$ is simply $P \setminus \{r_t\}$. Hence $\overline{D}$ and $\overline{D}'$ differ by exactly one record for user u^*, and therefore

$$d_{\text{rec}}(\overline{D}, \overline{D}') \leq 1.$$

So it remains to consider the case $k < m$. By maximality of k,

$$C_k \leq \Phi_{\text{user}}(u^*) < C_{k+1}. \tag{5}$$

For each $j \in \{0, \ldots, m\}$, define

$$P'_j := \{r_1, \ldots, r_j\} \setminus \{r_t\}.$$

Every prefix of the ordered records of u^* in D' can be written as P'_j for some j. Moreover,

$$\sum_{r \in P'_j} \Phi_{\text{rec}}(r) = \begin{cases} C_j, & j < t, \\ C_j - \Phi_{\text{rec}}(r_t), & j \geq t. \end{cases} \tag{6}$$

Let k' be the largest j such that P'_j is feasible, i.e.,

$$\sum_{r \in P'_j} \Phi_{\text{rec}}(r) \leq \Phi_{\text{user}}(u^*).$$

Then the retained prefix of user u^* in $\overline{D}'$ is

$$P' := P'_{k'}.$$

We now show that $k' \in \{k, k+1\}$.

First, P'_k is always feasible. If $t > k$, then $P'_k = P$, and its total privacy cost is $C_k \leq \Phi_{\text{user}}(u^*)$. If $t \leq k$, then

$$\sum_{r \in P'_k} \Phi_{\text{rec}}(r) = C_k - \Phi_{\text{rec}}(r_t) \leq C_k \leq \Phi_{\text{user}}(u^*).$$

Therefore,

$$k' \geq k. \tag{7}$$

Second, no P'_j with $j \geq k+2$ can be feasible. Indeed, if $j < t$, then $j > k$, so by (5),

$$\sum_{r \in P'_j} \Phi_{\text{rec}}(r) = C_j \geq C_{k+1} > \Phi_{\text{user}}(u^*).$$

If $j \geq t$, then by (3),

$$\Phi_{\text{rec}}(r_t) \leq \Phi_{\text{rec}}(r_j),$$

and hence

$$\begin{aligned} \sum_{r \in P'_j} \Phi_{\text{rec}}(r) &= C_j - \Phi_{\text{rec}}(r_t) \geq C_j - \Phi_{\text{rec}}(r_j) \\ &= C_{j-1} \geq C_{k+1} > \Phi_{\text{user}}(u^*). \end{aligned}$$

Thus,

$$k' \leq k+1. \tag{8}$$

Combining (7) and (8), we obtain

$$k' \in \{k, k+1\}.$$

Therefore, compared with

$$P = \{r_1, \ldots, r_k\},$$

the retained prefix P' is obtained by possibly removing the deleted record r_t, and possibly adding the single next record r_{k+1}. Equivalently, one can transform P into P' using at most one record deletion and at most one record insertion.

Since all users other than u^* have identical retained prefixes in $\overline{D}$ and $\overline{D}'$, it follows that

$$d_{\text{rec}}(\overline{D}, \overline{D}') \leq 2.$$

This completes the proof. □

A.2 Proof of Lemma 9

Proof. Fix any user u. We consider two cases.

First, suppose that $u \in D$. By the definition of prefix-truncation, for user u we retain the longest prefix of $D(u)$, under the privacy-based ordering, whose cumulative record-level privacy budget does not exceed $\Phi_{\text{user}}(u)$. Therefore, the retained records satisfy

$$\sum_{r \in \overline{D}(u)} \Phi_{\text{rec}}(r) \leq \Phi_{\text{user}}(u), \qquad \forall u \in D.$$

Otherwise, suppose that $u \notin D$. Then $D(u) = \emptyset$, and hence prefix-truncation also gives $\overline{D}(u) = \emptyset$. Thus,

$$\sum_{r \in \overline{D}(u)} \Phi_{\text{rec}}(r) = 0 < \Phi_{\text{user}}(u), \qquad \forall u \notin D.$$

Combining the two cases, for every user $u \in \mathcal{U}$,

$$\sum_{r \in \overline{D}(u)} \Phi_{\text{rec}}(r) \leq \Phi_{\text{user}}(u).$$

This completes the proof. □

A.3 Proof of Lemma 10

Proof. Since $D \sim_r D'$, without loss of generality, suppose that D' is obtained from D by adding one record r^* for some user u^*. The deletion case is symmetric.

Let D_τ and D'_τ be the datasets obtained from D and D' after the first truncation step. We consider two cases.

First, suppose that $\Phi_{\text{rec}}(r^*) < \tau$. Then r^* is removed in the first truncation step. Hence,

$$D_\tau = D'_\tau.$$

Otherwise, suppose that $\Phi_{\text{rec}}(r^*) \geq \tau$. Then r^* is retained. Since no other record is changed by the first truncation step, D_τ and D'_τ still differ by exactly one record. Thus,

$$d_{\text{rec}}(D_\tau, D'_\tau) \leq 1.$$

The privacy upgrade step only changes the record-level privacy budget of each retained record to τ and does not change the underlying record set. Therefore, after the first truncation and privacy upgrade, the two datasets remain identical or record-neighboring.

It remains to consider the final prefix-truncation step. After the privacy upgrade, every retained record has the same record-level privacy budget τ. Hence, sorting records by nondecreasing record-level privacy budget is equivalent to using the fixed natural order as the tie-breaking order.

For each user u, the number of retained records is

$$\kappa(u) = \min\left(\left\lfloor \frac{\Phi_{\text{user}}(u)}{\tau} \right\rfloor, |D_\tau(u)|\right),$$

which is exactly the length of the longest prefix whose cumulative upgraded record-level privacy budget is at most $\Phi_{\text{user}}(u)$.

Therefore, the second stage of doubly-truncation is precisely privacy-based ordering prefix-truncation applied to the datasets obtained after the first truncation and privacy upgrade. By Lemma 8, applying such prefix-truncation to two record-neighboring datasets can increase their record-level distance to at most 2. Consequently,

$$d_{\text{rec}}(\overline{\overline{D}}, \overline{\overline{D'}}) \leq 2.$$

This completes the proof. □

A.4 Proof of Lemma 11

Proof. Fix any user u. We consider two cases.

First, suppose that $u \in D$. By the definition of doubly-truncation, we first retain only the records

$$D_\tau(u) = \{r \in D(u) : \Phi_{\text{rec}}(r) \geq \tau\}.$$

For every record $r \in D_\tau(u)$, its privacy budget after doubly-truncation is upgraded to τ. Then, doubly-truncation retains the first $\kappa(u)$ records of $D_\tau(u)$, where

$$\kappa(u) = \min\left(\left\lfloor \frac{\Phi_{\text{user}}(u)}{\tau} \right\rfloor, |D_\tau(u)|\right).$$

Therefore, since every retained record has upgraded privacy budget τ, we have

$$\sum_{r \in \overline{\overline{D}}(u)} \tau = |\overline{\overline{D}}(u)| \cdot \tau = \kappa(u)\tau \leq \Phi_{\text{user}}(u), \qquad \forall u \in D.$$

Otherwise, suppose that $u \notin D$. Then $D(u) = \emptyset$, and hence $D_\tau(u) = \emptyset$. Therefore, doubly-truncation also gives $\overline{\overline{D}}(u) = \emptyset$. Thus,

$$\sum_{r \in \overline{\overline{D}}(u)} \tau = 0 \leq \Phi_{\text{user}}(u), \qquad \forall u \notin D.$$

Combining the two cases, for every user $u \in \mathcal{U}$,

$$\sum_{r \in \overline{\overline{D}}(u)} \tau \leq \Phi_{\text{user}}(u).$$

This completes the proof. □

A.5 Proof of Lemma 12

Proof. By symmetry, it suffices to consider the case where D' is obtained from D by deleting one record r^*.

Fix τ_i. In the first truncation step, records with $\Phi_{\text{rec}}(r) < \tau_i$ are removed independently record by record. Hence, after this step, the resulting datasets are either identical or still record-neighboring.

If they are identical, then applying the second step yields

$$\overline{\overline{D}}_i = \overline{\overline{D'}}_i,$$

and thus

$$|S_i(D) - S_i(D')| = 0.$$

Otherwise, the two intermediate datasets are record-neighboring. Applying prefix-truncation in the second step, the argument in the proof of Lemma 8 implies that deleting one record can remove that record from the retained prefix and admit at most one additional record. Therefore,

$$\left||\overline{\overline{D}}_i| - |\overline{\overline{D'}}_i|\right| \leq 1.$$

Since

$$S_i(D) = |\overline{\overline{D}}_i|, \qquad S_i(D') = |\overline{\overline{D'}}_i|,$$

we obtain

$$|S_i(D) - S_i(D')| \leq 1.$$

This completes the proof. □

A.6 Proof of Theorem 5.1

Proof. Recall that Algorithm 2 sets

$$\ell = \lceil \log(\hat{\varepsilon}/\check{\varepsilon}) \rceil + 1, \quad \tau_i = 2^{i-1}\check{\varepsilon},$$

and releases

$$\widetilde{S_i} = \left|\overline{\overline{D}}_i\right| - \frac{3\ell}{\tau_i} \ln\frac{\ell}{\beta} + \text{Lap}\left(\frac{\ell}{\tau_i}\right).$$

Privacy. We prove that Algorithm 2 is $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP. Recall that, for each iteration i, S_i is the non-noisy score computed after doubly-truncation with threshold τ_i. And the released privatized score is

$$\widetilde{S_i} = S_i - \frac{3\ell}{\tau_i} \ln\frac{\ell}{\beta} + \text{Lap}\left(\frac{\ell}{\tau_i}\right).$$

The deterministic bias term does not affect privacy.

Record-level privacy guarantee for each iteration. We first consider the record-level guarantee for a fixed iteration i. Fix any record r. If $\tau_i > \Phi_{\text{rec}}(r)$, then by the record-level truncation in doubly-truncation, record r is removed before forming the truncated dataset. Hence the distribution of $\widetilde{S_i}$ is identical whether or not r is present, and the privacy loss for r in this iteration is **zero**.

Otherwise, if $\tau_i \le \Phi_{\text{rec}}(r)$, then by Lemma 12, the record-level sensitivity of S_i is at most one, i.e., $\text{GS}_{S_i} = 1$.

Therefore, by the Laplace mechanism, adding $\text{Lap}(\ell/\tau_i)$ ensures that releasing $\widetilde{S}_i$ satisfies (τ_i/ℓ)-record-level DP with respect to record r. Since $\tau_i \le \Phi_{\text{rec}}(r)$, we have $\tau_i/\ell \le \Phi_{\text{rec}}(r)/\ell$. Thus, the i-th iteration satisfies (Φ_{rec}/ℓ)-record-level privacy guarantee.

User-level privacy guarantee for each iteration. We next verify the user-level guarantee for the same fixed iteration i. Fix any user u, and consider two user-neighboring datasets that differ only in the records contributed by user u. Recall that $\overline{\overline{D}}_i(u)$ is the retained records of user u after doubly-truncation with threshold τ_i. Since the score S_i is the size of the truncated dataset, the change in S_i caused by adding or removing user u is at most $|\overline{\overline{D}}_i(u)|$. Therefore, releasing $\widetilde{S}_i$ with Laplace noise scale ℓ/τ_i incurs user-level privacy loss at most

$$\frac{|\overline{\overline{D}}_i(u)|}{\ell/\tau_i} = \frac{1}{\ell} \sum_{r \in \overline{\overline{D}}_i(u)} \tau_i.$$

By Lemma 11, doubly-truncation guarantees that $\sum_{r \in \overline{\overline{D}}_i(u)} \tau_i \le \Phi_{\text{user}}(u)$. Hence the privacy loss for user u in the i-th iteration is at most $\Phi_{\text{user}}(u)/\ell$. Thus, the i-th iteration satisfies the $(\Phi_{\text{user}}/\ell)$-user-level privacy guarantee.

Composition across iterations. Combining this with the record-level argument above, each iteration is $(\Phi_{\text{rec}}/\ell, \Phi_{\text{user}}/\ell)$ -Hybrid-PDP. Applying the sequential composition (Lemma 6) over the ℓ iterations, the release of the full scores $(\widetilde{S}_1, \ldots, \widetilde{S}_\ell)$ satisfies $(\Phi_{\text{rec}}, \Phi_{\text{user}})$ -Hybrid-PDP. Finally, the selected index $i^\star = \arg\max_{1 \le i \le \ell} \widetilde{S}_i$ and the output threshold $\tau_{i^\star}$ are post-processing of the released scores. Therefore, by Lemma 5, releasing $\tau_{i^\star}$ also satisfies $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP.

Utility. By the Laplace tail bound in Lemma 4, for each iteration i, with probability $1 - \beta/\ell$,

$$-\frac{\ell}{\tau_i} \ln\frac{\ell}{\beta} \;\le\; \text{Lap}\left(\frac{\ell}{\tau_i}\right) \;\le\; \frac{\ell}{\tau_i} \ln\frac{\ell}{\beta}. \tag{9}$$

By applying the union bound over iterations $i = 1, 2, \ldots, \ell$, with probability at least $1 - \beta$, the following holds:

$$-\frac{\ell}{\tau_i} \ln\frac{\ell}{\beta} \;\le\; \text{Lap}\left(\frac{\ell}{\tau_i}\right) \;\le\; \frac{\ell}{\tau_i} \ln\frac{\ell}{\beta}, \quad i = 1, 2, \ldots, \ell. \tag{10}$$

With (10), we prove claims (i) and (ii) separately.

Recall (i) $\tau_{i^\star} \ge \frac{1}{2}\,\varepsilon_{\min}(D)$:

Let k be the index such that

$$\varepsilon_{\min}(D) \in [\tau_k,\, 2\tau_k).$$

By the definition of $\varepsilon_{\min}(D)$, for every $j \le k$, no record is removed by doubly-truncation. Hence

$$\left|\overline{\overline{D}}_j\right| = |D| \quad \forall j \le k.$$

Fix any $j \le k-1$. Using the above and expanding $\widetilde{S}_k - \widetilde{S}_j$, we have

$$\begin{aligned}
\widetilde{S}_k - \widetilde{S}_j &= \left|\overline{\overline{D}}_k\right| - \left|\overline{\overline{D}}_j\right| - \left(\frac{3\ell}{\tau_k} - \frac{3\ell}{\tau_j}\right)\ln\frac{\ell}{\beta} + \text{Lap}\left(\frac{\ell}{\tau_k}\right) - \text{Lap}\left(\frac{\ell}{\tau_j}\right) \\
&= -\left(\frac{3\ell}{\tau_k} - \frac{3\ell}{\tau_j}\right)\ln\frac{\ell}{\beta} + \text{Lap}\left(\frac{\ell}{\tau_k}\right) - \text{Lap}\left(\frac{\ell}{\tau_j}\right) \\
&\ge -\left(\frac{3\ell}{\tau_k} - \frac{3\ell}{\tau_j}\right)\ln\frac{\ell}{\beta} - \left(\frac{\ell}{\tau_k} + \frac{\ell}{\tau_j}\right)\ln\frac{\ell}{\beta} \\
&= \left(\frac{2\ell}{\tau_j} - \frac{4\ell}{\tau_k}\right)\ln\frac{\ell}{\beta}.
\end{aligned}$$

Since $j \le k-1$ implies $\tau_j \le \tau_k/2$, we have $1/\tau_j \ge 2/\tau_k$, and thus

$$\left(\frac{2\ell}{\tau_j} - \frac{4\ell}{\tau_k}\right)\ln\frac{\ell}{\beta} \ge 0.$$

Therefore, $\widetilde{S}_k \ge \widetilde{S}_j$ for all $j \le k-1$. That is, no index $j < k$ has a larger privatized score than index k. Therefore,

$$\tau_{i^\star} \ge \tau_k > \tfrac{1}{2}\,\varepsilon_{\min}(D),$$

which proves claim (i).

Recall (ii) $\left|\,|D| - |\overline{\overline{D}}_{i^\star}|\right| = O\left(\frac{1}{\varepsilon_{\min}(D)} \log(\hat{\varepsilon}/\check{\varepsilon}) \log\frac{\log(\hat{\varepsilon}/\check{\varepsilon})}{\beta}\right)$, where $\overline{\overline{D}}_{i^\star}$ is obtained by applying doubly-truncation to D with threshold $\tau_{i^\star}$:

Since $i^\star$ is the index with the best score,$S_{i^\star}$, we have

$$\widetilde{S}_{i^\star} \ge \widetilde{S}_k.$$

Expanding both $\widetilde{S}_{i^\star}$ and $\widetilde{S}_k$, and using $\left|\overline{\overline{D}}_k\right| = |D|$, we obtain

$$\left|\overline{\overline{D}}_{i^\star}\right| - \frac{3\ell}{\tau_{i^\star}}\ln\frac{\ell}{\beta} + \text{Lap}\left(\frac{\ell}{\tau_{i^\star}}\right) \ge |D| - \frac{3\ell}{\tau_k}\ln\frac{\ell}{\beta} + \text{Lap}\left(\frac{\ell}{\tau_k}\right).$$

Rearranging yields

$$|D| - \left|\overline{\overline{D}}_{i^\star}\right| \le \text{Lap}\left(\frac{\ell}{\tau_{i^\star}}\right) - \text{Lap}\left(\frac{\ell}{\tau_k}\right) + \left(\frac{3\ell}{\tau_k} - \frac{3\ell}{\tau_{i^\star}}\right)\ln\frac{\ell}{\beta}.$$

Applying (10) to both Laplace terms gives

$$\begin{aligned}
|D| - \left|\overline{\overline{D}}_{i^\star}\right| &\le \left(\frac{\ell}{\tau_{i^\star}} + \frac{\ell}{\tau_k}\right)\ln\frac{\ell}{\beta} + \left(\frac{3\ell}{\tau_k} - \frac{3\ell}{\tau_{i^\star}}\right)\ln\frac{\ell}{\beta} \\
&= \left(\frac{4\ell}{\tau_k} - \frac{2\ell}{\tau_{i^\star}}\right)\ln\frac{\ell}{\beta} \\
&\le \frac{4\ell}{\tau_k}\ln\frac{\ell}{\beta}.
\end{aligned}$$

Finally, since $\varepsilon_{\min}(D) \in [\tau_k, 2\tau_k)$, we have $1/\tau_k \le 2/\varepsilon_{\min}(D)$, and thus

$$|D| - \left|\overline{\overline{D}}_{i^\star}\right| \le \frac{8\ell}{\varepsilon_{\min}(D)}\ln\frac{\ell}{\beta}.$$

Noting that $\ell = \lceil \log(\hat{\varepsilon}/\check{\varepsilon}) \rceil + 1$, this implies

$$|D| - \left|\overline{\overline{D}}_{i^\star}\right| = O\left(\frac{\ell}{\varepsilon_{\min}(D)}\log\frac{\ell}{\beta}\right) = O\left(\frac{1}{\varepsilon_{\min}(D)}\log(\hat{\varepsilon}/\check{\varepsilon})\log\frac{\log(\hat{\varepsilon}/\check{\varepsilon})}{\beta}\right),$$

which proves claim (ii). □

A.7 Proof of Theorem 5.2

Proof. We prove the theorem in two parts: privacy and utility.

Privacy. We first prove that Algorithm 3 satisfies $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP.

(1) In Stage 1, the algorithm invokes

$$\textsc{ThresholdSelection}(D, \Phi_{\text{rec}}/2, \Phi_{\text{user}}/2, \beta/2).$$

By Theorem 5.1, this step satisfies $(\Phi_{\text{rec}}/2, \Phi_{\text{user}}/2)$-Hybrid-PDP. The subsequent assignment

$$\tau_{i^\star} = 2\tau_{i^\star}^{\text{half}}$$

is deterministic post-processing of the selected threshold and therefore does not consume any additional privacy budget.

(2) In Stage 2, the algorithm computes

$$\overline{\overline{D}}_{i^\star} \leftarrow \textsc{Doubly-Truncation}(D, \Phi_{\text{rec}}, \Phi_{\text{user}}, \tau_{i^\star})$$

and then invokes

$$\mathcal{M}(\overline{\overline{D}}_{i^\star}, \tau_{i^\star}/4, \beta/2).$$

Record-level guarantee in Stage 2. For any record r, if $\Phi_{\text{rec}}(r) < \tau_{i^\star}$, then r is discarded by doubly-truncation and consumes no privacy budget in Stage 2. Otherwise, $\Phi_{\text{rec}}(r) \geq \tau_{i^\star}$. By Lemma 10, after doubly-truncation, the distance between two record-level neighboring datasets is at most 2. Since $\mathcal{M}$ is invoked with privacy budget $\tau_{i^\star}/4$, by group privacy (Lemma 1), the privacy consumption of any record r in Stage 2 is at most

$$2 \cdot \frac{\tau_{i^\star}}{4} = \frac{\tau_{i^\star}}{2} \leq \frac{\Phi_{\text{rec}}(r)}{2}.$$

Hence Stage 2 satisfies the $\Phi_{\text{rec}}/2$ record-level privacy requirement.

User-level guarantee for Stage 2. By Lemma 11, after doubly-truncation with threshold $\tau_{i^\star}$, the retained records of any user u have total upgraded record-level budget at most $\Phi_{\text{user}}(u)$. Since each retained record is assigned privacy budget $\tau_{i^\star}$, after doubly-truncation, two user-level neighboring datasets that differ in user u can differ in at most

$$2 \cdot \frac{\Phi_{\text{user}}(u)}{\tau_{i^\star}}$$

retained records. By group privacy (Lemma 1), invoking $\mathcal{M}$ with budget $\tau_{i^\star}/4$ therefore incurs user-level privacy loss at most

$$2 \cdot \frac{\Phi_{\text{user}}(u)}{\tau_{i^\star}} \cdot \frac{\tau_{i^\star}}{4} = \frac{\Phi_{\text{user}}(u)}{2}.$$

Thus Stage 2 satisfies the $\Phi_{\text{user}}/2$ user-level privacy requirement.

Therefore, Stage 2 is $(\Phi_{\text{rec}}/2, \Phi_{\text{user}}/2)$-Hybrid-PDP.

By the sequential composition of Hybrid-PDP (Lemma 6), the whole algorithm satisfies $(\Phi_{\text{rec}}, \Phi_{\text{user}})$-Hybrid-PDP.

Utility. Let D' denote the selected retained dataset $\overline{\overline{D}}_{i^\star}$ in Algorithm 3. Since the threshold selection in our framework runs with privacy functions $\Phi_{\text{rec}}/2$ and $\Phi_{\text{user}}/2$ and failure rate $\beta/2$ Theorem 5.1 implies that, with probability at least $1 - \beta/2$, the selected threshold $\tau_{i^\star}^{\text{half}}$ satisfies

$$\tau_{i^\star}^{\text{half}} \geq \frac{1}{2} \cdot \frac{\varepsilon_{\min}(D)}{2} = \frac{\varepsilon_{\min}(D)}{4}.$$

Since Algorithm 3 sets $\tau_{i^\star} = 2\tau_{i^\star}^{\text{half}}$, we have

$$\tau_{i^\star} \geq \frac{1}{2}\varepsilon_{\min}(D).$$

Moreover, the same theorem gives

$$\big||D| - |D'|\big| = O\left(\frac{1}{\varepsilon_{\min}(D)} \log(\hat{\varepsilon}/\check{\varepsilon}) \log\frac{\log(\hat{\varepsilon}/\check{\varepsilon})}{\beta}\right). \tag{11}$$

With D', Algorithm 3 runs $\mathcal{M}$ with privacy budget $\tau_{i^\star}/4$ and failure probability $\beta/2$ to output $\widetilde{Q}(D)$. Since

$$\frac{\tau_{i^\star}}{4} \geq \frac{\varepsilon_{\min}(D)}{8},$$

by the definition of $\text{Err}_{\mathcal{M}}(\cdot)$, with probability at least $1 - \beta/2$,

$$\left|\widetilde{Q}(D) - Q(D')\right| \leq \text{Err}_{\mathcal{M}}\big(D',\ \varepsilon_{\min}(D)/8,\ \beta/2\big). \tag{12}$$

By a union bound over (11) and (12), both hold simultaneously with probability at least $1 - \beta$. By the triangle inequality,

$$\begin{aligned}\left|\widetilde{Q}(D) - Q(D)\right| &\leq \left|Q(D) - Q(D')\right| + \left|\widetilde{Q}(D) - Q(D')\right| \\ &\leq \left|Q(D) - Q(D')\right| + \text{Err}_{\mathcal{M}}\big(D',\ \varepsilon_{\min}(D)/8,\ \beta/2\big),\end{aligned}$$

where D' is obtained from D by removing

$$O\left(\frac{1}{\varepsilon_{\min}(D)} \log(\hat{\varepsilon}/\check{\varepsilon}) \log\frac{\log(\hat{\varepsilon}/\check{\varepsilon})}{\beta}\right)$$

records. □

B DATASET PREPROCESSING TECHNIQUES AND SYNTHETIC DATASET DETAILS

B.1 Dataset Preprocessing and Definitions

Dataset Name	User Identifying Attributes	Value Attributes
CK Pay	Bureau, employee	Quarterly salary (USD)
NY Salary	Authority, employee	Annual salary (USD)
VLE	Student	Sum click (Click)
Online Retail	Invoice	Price × quantity (GBP)
Czech Bank	Account	Amount (CZK)
TPCH Order	Customer	Total price
TPCH Lineitem	Supplier	Extend price

Table 4: Definitions of users, records, and values across datasets.

Preprocessing. For all raw datasets, we first remove (i) rows with missing target values and (ii) rows whose user-identifying attributes are missing. We clip the value of records in all datasets to the range $[0, 2^{30}]$. Using the integer domain is not a limitation but a standard choice in the DP literature, since fractional values can be quantized to integers.

Definitions of users, records, and values. In our experiments, each retained row after preprocessing corresponds to a *record*. The *value* of a record is the numeric attribute used by the query. Meanwhile, a *user* is defined as the collection of records that share the same set of user-identifying attributes in the retained dataset. The concrete definitions of record value attributes and user-identifying attributes for each dataset are summarized in Table 4.

Category	Dataset Name	User Group	Record Group	Parameter
Normal	CK Pay	Bureau	Season	$\alpha = -2$
	NY Salary	Authority	Month	
	VLE	Student	Site	
Linear	Online Retail	Invoice	No record group	$\theta = -5$
	Czech Bank	Account		$\theta = -12$
Sqrt	TPCH Order	Area		$\gamma = -12$
	TPCH Lineitem	Supplier		$\gamma = -12$

Table 5: Grouping schemes and dataset-specific parameters for personalized privacy budgets.

B.2 Privacy Specification

User-level budgets. For each dataset, we partition users, as defined in Table 4, into disjoint groups according to the *user group* attribute listed in Table 5. Users within the same group share the same privacy budget, while the budgets of different groups are independently sampled from a distribution, aligning with common real-world privacy requirements. Specifically, for each user group, we sample a group-level budget $\varepsilon = 2^2 \cdot 2^{\mathcal{N}(0,\sigma)}$ with $\sigma = 0.5$, and assign it to every user u in the group. To enforce the global bounds, if the sampled budget falls outside $[\check{\varepsilon}_{\text{user}}, \hat{\varepsilon}_{\text{user}}]$, it is discarded and resampled until it satisfies the constraints.

Record-level budgets. Record-level privacy budgets are generated under three specifications, namely *normal*, *linear*, and *sqrt*, in order to capture the heterogeneity of record-level privacy requirements in practice, where the budgets can be independent of the record value or scale with the record value. Recall that $F(r)$ denotes the value associated with record r as defined in Section B.1. Throughout, we use $2^{\mathcal{N}(0,\sigma)}$ with $\sigma = 0.5$ to reflect various privacy needs. Details of each privacy specification are listed below:

- *Normal.* We partition records into disjoint record groups, and all records within the same group share a common record-level budget. For each record group, we sample $\Phi_{\text{rec}}(r) = 2^{\alpha} \cdot 2^{\mathcal{N}(0,\sigma)}$, where α is fixed to -2 for all datasets under the normal specification.
- *Linear.* We assign each record r a budget proportional to its value: $\Phi_{\text{rec}}(r) = 2^{\theta} \cdot 2^{\mathcal{N}(0,\sigma)} \cdot F(r)$, where θ is dataset-specific as reported in Table 5.
- *Sqrt.* We assign each record r a budget proportional to the square root of its value: $\Phi_{\text{rec}}(r) = 2^{\gamma} \cdot 2^{\mathcal{N}(0,\sigma)} \cdot \big(F(r)\big)^{1/2}$, where γ is dataset-specific as reported in Table 5.

Similar to the user-level case, to enforce global constraints we require $\Phi_{\text{rec}}(r) \in [\check{\varepsilon}_{\text{rec}}, \hat{\varepsilon}_{\text{rec}}]$ for every record r.